

Distance and resistance on random series–parallel graphs: logarithmic speeds and near-critical asymptotics

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ABSTRACT

We study the graph distance $D_n(p)$ and effective resistance $R_n(p)$ between the boundary vertices of a depth- n random series–parallel graph, obtained by recursively joining two independent copies in series with probability p and in parallel with probability $1 - p$. We prove the existence of deterministic logarithmic speeds: for every $p \in [0, 1]$, $n^{-1} \log D_n(p)$ and $n^{-1} \log R_n(p)$ converge to deterministic limits $v_D(p)$ and $v_R(p)$, respectively, almost surely and in L^1 . The limiting speeds agree with the corresponding first-moment logarithmic rates for every $p \in [0, 1]$ in the distance case and for $p \in [1/2, 1]$ in the resistance case. We further determine the near-critical behavior of the resistance speed: $v_R(\frac{1}{2} + \delta) \sim 2\zeta(3)^{1/3} \lambda_* \delta^{2/3}$ as $\delta \downarrow 0$, where $\lambda_* > 0$ is characterized by an explicit nonlinear boundary-value problem. This exponent $2/3$ contrasts with the exponent $1/2$ for distance obtained by Chen, Derrida, Duquesne, and Shi (2026). A companion Lean 4 development formalizes the three principal theorems and their proof graph, subject to two explicitly documented external mathematical assumptions.

1. Introduction

Hierarchical random systems provide a natural setting for studying the competition between recursive mechanisms acting in opposite directions. Their balance may give rise to a critical regime, while a small bias away from balance can be amplified across scales. A natural question is then how the system departs from its critical behavior when the balance is weakly broken, and how the scale of this departure depends on the strength of the bias.

The random series–parallel graph introduced by Hambly and Jordan [10] provides a concrete instance of this phenomenon. Starting from a single edge, at each generation every edge is independently replaced by two edges in series with probability p , and by two edges in parallel with probability $1 - p$. Let $D_n(p)$ and $R_n(p)$ denote, respectively, the graph distance and the effective resistance between the two boundary vertices after n generations. Hambly and Jordan [10, Theorem 2.1 and Lemma 3.1] proved that, almost surely, both $D_n(p)$ and $R_n(p)$ tend to infinity when $p > \frac{1}{2}$, whereas when $p < \frac{1}{2}$, $D_n(p)$ converges to a finite limit and $R_n(p)$ tends to zero.

These phase-transition results describe only the qualitative long-time behavior of $D_n(p)$ and $R_n(p)$. Since the same random operation is iterated at each generation, it is natural to ask whether their logarithms are asymptotically linear in the generation number. This leads to the following questions:

1. For every fixed $p \in [0, 1]$, do $n^{-1} \log D_n(p)$ and $n^{-1} \log R_n(p)$ have deterministic almost-sure limits? Do $n^{-1} \log \mathbb{E} D_n(p)$ and $n^{-1} \log \mathbb{E} R_n(p)$ also converge? When the relevant limits exist, how are the almost-sure logarithmic speeds and the first-moment logarithmic rates related?
2. If these logarithmic speeds exist, what are their near-critical asymptotics as p approaches the critical value $\frac{1}{2}$?

For the first problem, Hambly and Jordan [10, Lemma 3.1(a)] established the almost-sure exponential growth of $D_n(p)$ for $p > 1/2$. Chen, Derrida, Duquesne, and Shi [7, equation (1.2)] later proved that the first-moment logarithmic rate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} D_n(p)$$

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exists for every $p \in [0, 1]$ and is positive exactly when $p > 1/2$. The latter work did not establish the corresponding almost-sure convergence for all p . For resistance, Hambly and Jordan obtained almost-sure exponential upper and lower bounds for $p \neq 1/2$ and proved that $n^{-1} \log R_n(1/2) \rightarrow 0$ almost surely, but left the general convergence question open.

It is also worth noting that, at criticality, more precise sublinear behavior is known for both quantities. Auffinger and Cable [5, Theorem 1] proved that $\log D_n(1/2)$ has scale $\sqrt{n}$ and identified its limiting law. For resistance, Addario-Berry et al. [1] predicted an $n^{1/3}$ -scale limit law for $\log R_n(1/2)$; this was recently proved independently by Chen, Duquesne, and Shi [8] and Morfe [14].

For the second problem, Chen, Derrida, Duquesne, and Shi [7, Theorem 1] obtained the sharp near-critical asymptotics of this first-moment logarithmic rate for distance, showing that it vanishes on the scale $(p - \frac{1}{2})^{1/2}$ as $p \downarrow \frac{1}{2}$. No corresponding near-critical asymptotic for the resistance logarithmic speed was known.

Combined with these earlier results, our work answers the two questions above. We prove that, for every $p \in [0, 1]$, $n^{-1} \log D_n(p)$ and $n^{-1} \log R_n(p)$ converge almost surely and in L^1 to deterministic limits. We also prove that $n^{-1} \log \mathbb{E} D_n(p)$ and $n^{-1} \log \mathbb{E} R_n(p)$ converge. The distance rate agrees with $v_D(p)$ throughout $[0, 1]$, while the resistance rate is $v_R(p) \vee \log(2p)$ for $0 \leq p \leq 1$, with the maximum equal to $v_R(p)$ for $1/2 \leq p \leq 1$, and to $\log(2p)$ for $p < 1/2$ sufficiently close to $1/2$. We further determine the sharp near-critical asymptotics of the resistance speed, showing that its magnitude vanishes on the scale $|p - \frac{1}{2}|^{2/3}$ as $p \rightarrow \frac{1}{2}$.

1.1. Main results

We define the random series-parallel graph through a binary-tree construction. Let

$$\mathbb{T} := \bigcup_{k \geq 0} \{0, 1\}^k, \quad \{0, 1\}^0 := \{\emptyset\},$$

and let $(\xi_u)_{u \in \mathbb{T}}$ be independent Bernoulli random variables with parameter p . The initial graph $\mathcal{G}_0(p)$ consists of a single edge $e_\emptyset$ joining two distinguished vertices s and t , called the boundary vertices. Suppose recursively that the edges of $\mathcal{G}_k(p)$ are indexed by $u \in \{0, 1\}^k$. For each edge e_u with endpoints x_u and y_u , perform the following replacement. If $\xi_u = 1$, introduce a new vertex z_u and replace e_u by the two edges

$$e_{u0} = \{x_u, z_u\}, \quad e_{u1} = \{z_u, y_u\}$$

in series. If $\xi_u = 0$, replace e_u by two distinct parallel edges e_{u0} and e_{u1} , both joining x_u and y_u . Performing these replacements simultaneously for all $u \in \{0, 1\}^k$ produces $\mathcal{G}_{k+1}(p)$. Figure 1 illustrates the first two refinement steps for one realization of the environment.

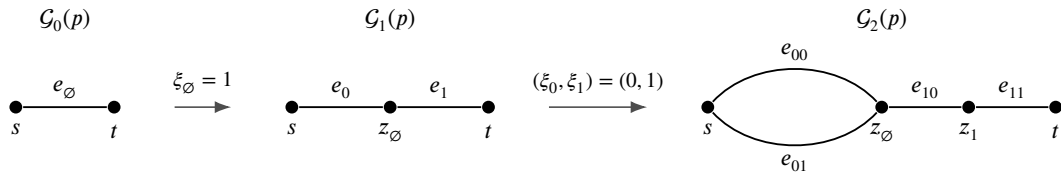


Figure 1: Two successive refinement steps in one realization of the environment. The choice $\xi_\emptyset = 1$ replaces the initial edge in series. At the next level, $\xi_0 = 0$ replaces e_0 in parallel, while $\xi_1 = 1$ replaces e_1 in series.

At every level, s and t remain the distinguished boundary vertices. Let $D_n(p)$ be the graph distance between s and t in $\mathcal{G}_n(p)$, and let $R_n(p)$ be their effective resistance when every edge has unit resistance.

Theorem 1.1 (Logarithmic speeds). *For every $p \in [0, 1]$, there exist deterministic constants $v_D(p)$ and $v_R(p)$ such that*

$$\frac{1}{n} \log D_n(p) \longrightarrow v_D(p), \quad \frac{1}{n} \log R_n(p) \longrightarrow v_R(p) \quad \text{a.s. and in } L^1. \quad (1.1)$$

Their values satisfy

$$v_D(p) = 0 \quad \text{for } 0 \leq p \leq \frac{1}{2}, \quad v_R(1-p) = -v_R(p) \quad \text{for } 0 \leq p \leq 1,$$

so in particular $v_R(1/2) = 0$. Moreover, for $1/2 < p \leq 1$,

$$v_D(p), v_R(p) \in [\log(2p), \log 2]. \quad (1.2)$$

Theorem 1.2 (First-moment logarithmic rates). *For every $p \in [0, 1]$, the limits*

$$\gamma_D(p) := \lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} D_n(p), \quad \gamma_R(p) := \lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} R_n(p)$$

exist. With the convention $\log 0 = -\infty$, they are related to the logarithmic speeds by

$$\gamma_D(p) = v_D(p), \quad \gamma_R(p) = v_R(p) \vee \log(2p) \quad \text{for } 0 \leq p \leq 1. \quad (1.3)$$

In particular,

$$\gamma_R(p) = v_R(p) \quad \text{for } \frac{1}{2} \leq p \leq 1. \quad (1.4)$$

Figure 2 schematically illustrates the maximum in (1.3).

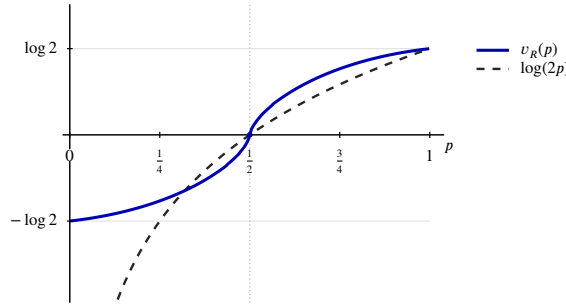


Figure 2: Schematic comparison of the two terms in $\gamma_R(p) = v_R(p) \vee \log(2p)$. For $1/2 \leq p \leq 1$, Theorem 1.2 gives $v_R(p) \geq \log(2p)$, so the maximum is $v_R(p)$. For $p < 1/2$ sufficiently close to $1/2$, Theorem 1.3 gives $\log(2p) > v_R(p)$, so the maximum is $\log(2p)$. No claim is made about the global curvature of v_R or the number or locations of crossings between $v_R(p)$ and $\log(2p)$ in $0 < p < 1/2$. The plotted global continuation is arbitrary and is not computed data.

Theorem 1.3 (Resistance speed near criticality). *For $\lambda > 0$, consider the boundary-value problem*

$$\begin{aligned} W^2 W' - \lambda W + u(1-u) &= 0, & 0 < u < 1, \\ W(u) > 0 & \text{ for } 0 < u < 1, & W(0) = W(1) = 0. \end{aligned} \quad (1.5)$$

The set of $\lambda > 0$ for which this problem admits a solution $W \in C([0, 1]) \cap C^1((0, 1))$ has a smallest element $\lambda_ \in (0, \infty)$. Moreover, as $\delta \downarrow 0$, we have*

$$v_R\left(\frac{1}{2} + \delta\right) = -v_R\left(\frac{1}{2} - \delta\right) \sim 2\zeta(3)^{1/3} \lambda_* \delta^{2/3}, \quad (1.6)$$

and

$$\gamma_R\left(\frac{1}{2} + \delta\right) \sim 2\zeta(3)^{1/3} \lambda_* \delta^{2/3}, \quad \gamma_R\left(\frac{1}{2} - \delta\right) \sim -2\delta. \quad (1.7)$$

Remark 1.4. For comparison, Chen, Derrida, Duquesne, and Shi [7, equation (1.2) and Theorem 1] proved that the first-moment logarithmic rate for distance exists and obtained its sharp near-critical asymptotics. Together with Theorem 1.2, their results give

$$v_D\left(\frac{1}{2} + \delta\right) = \gamma_D\left(\frac{1}{2} + \delta\right) \sim \sqrt{\zeta(2)} \delta^{1/2} = \frac{\pi}{\sqrt{6}} \delta^{1/2} \quad \text{as } \delta \downarrow 0.$$

Thus the distance and resistance speeds vanish with different critical exponents $1/2$ and $2/3$, respectively. Remark 4.8 explains heuristically why the resistance exponent is $2/3$.

1.2. Proof outline

We next outline the main ideas behind the proofs of Theorems 1.1–1.3.

To prove Theorems 1.1 and 1.2, we first establish the existence of the first-moment logarithmic rates. The argument uses the conditional refinement property in Proposition 2.2: conditional on $\mathcal{G}_n(p)$, the network $\mathcal{G}_{n+r}(p)$ is obtained by replacing each edge of $\mathcal{G}_n(p)$ with an independent copy of $\mathcal{G}_r(p)$. For distance, we replace each edge of a shortest path in $\mathcal{G}_n(p)$ by a shortest path across the corresponding depth- r copy; the resulting candidate path has conditional expected length $(\mathbb{E}D_r(p))D_n(p)$. For resistance, let (θ_e) be a minimizing unit flow on $\mathcal{G}_n(p)$ and, in the copy replacing e , route θ_e times a minimizing unit flow. The resulting candidate flow has conditional expected energy $(\mathbb{E}R_r(p))R_n(p)$. Taking expectations gives first-moment submultiplicativity, and Fekete's subadditive lemma then gives the desired rates. See Lemma 3.1 for details.

The main remaining issue is to show that, for both $D_n(p)$ and $R_n(p)$, the gap between the logarithm of the mean and the mean logarithm, which we call the Jensen gap, is $o(n)$. For $p > 1/2$, on the respective events $\{D_n(p) \geq \mathbb{E}D_n(p)/2\}$ and $\{R_n(p) \geq \mathbb{E}R_n(p)/2\}$, the corresponding Jensen gaps are bounded by $\log 2$ plus the absolute fluctuations of the logarithms around their means, as shown in (3.10). Lemma 3.3 bounds the expectations of these fluctuations uniformly in n . The two events have probabilities uniformly bounded away from zero by the second-moment method; see Lemma 3.4 and (3.9). This identifies the first-moment logarithmic rates with the logarithmic speeds. The fluctuation control in Lemma 3.3 upgrades convergence of the mean logarithms to the almost-sure and L^1 convergence in (1.1).

To prove Theorem 1.3, we construct two families of random variables that yield upper and lower bounds for $\log R_n(\frac{1}{2} + \delta)$. Consider the upper bound, for example. Let $\mathcal{T}_{\frac{1}{2}+\delta}$ denote the one-step CDF update of $\log R_n(\frac{1}{2} + \delta)$ (see Section 2.3), and write $\delta = \varepsilon^3$. For any $\lambda > \lambda_*$ and $\kappa > \kappa_\lambda := 2\zeta(3)^{1/3}\lambda$, and all sufficiently small ε , we construct a random variable Y_ε with CDF G_ε such that

$$F_0 \geq G_\varepsilon, \quad \mathcal{T}_{1/2+\varepsilon^3}G_\varepsilon(x) \geq G_\varepsilon(x - \kappa\varepsilon^2). \quad (1.8)$$

Here $F_0(x) = \mathbf{1}_{\{x \geq 0\}}$ is the CDF of $\log R_0(\frac{1}{2} + \varepsilon^3)$. The first inequality in (1.8) gives $\log R_0(\frac{1}{2} + \varepsilon^3) \leq_{\text{st}} Y_\varepsilon$. Then the second inequality in (1.8), combined with the basic properties of random series-parallel graphs (see Lemma 2.6), gives $\log R_n(\frac{1}{2} + \varepsilon^3) \leq_{\text{st}} Y_\varepsilon + n\kappa\varepsilon^2$, and hence $\nu_R(1/2 + \varepsilon^3) \leq \kappa\varepsilon^2$. Letting $\kappa \downarrow \kappa_\lambda$ and then $\lambda \downarrow \lambda_*$ gives the sharp upper bound. The matching lower bound follows from the same reasoning.

To construct G_ε satisfying (1.8), we compare the one-step change $\mathcal{T}_{1/2+\delta}G_\varepsilon(x) - G_\varepsilon(x)$ with the change $G_\varepsilon(x - \kappa\varepsilon^2) - G_\varepsilon(x)$ caused by translation. Matching their leading-order coefficients yields the equation in Theorem 1.3:

$$W^2W' - \lambda W + u(1 - u) = 0.$$

The ODE analysis ensures the existence of its solution, thus ensuring the existence of G_ε .

Remark 1.5. The differential equation in Theorem 1.3 comes directly from the above comparison between the CDF update and the translation. Test the operator on a slowly varying CDF $F_{\varepsilon, z_0}(x) = \Phi(z_0 + \varepsilon x)$, with $q = \Phi'$. Lemma 2.6 and Lemma 4.4 give, for all x and z_0 ,

$$\mathcal{T}_{1/2+\varepsilon^3}F_{\varepsilon, z_0}(x) - F_{\varepsilon, z_0}(x) = \varepsilon^3(aqq' - 2\Phi(1 - \Phi)) + O(\varepsilon^5q).$$

Here the quantities on the right are evaluated at $z_0 + \varepsilon x$. A translation by $\kappa\varepsilon^2$ changes the same CDF by $-\kappa\varepsilon^3q + O(\varepsilon^6q)$. Matching the two leading terms therefore yields the equation

$$aq q' - 2\Phi(1 - \Phi) = -\kappa q.$$

With $q = \beta W(1 - \Phi)$, $\beta^3 = 2/a$, and $\kappa = 2\lambda/\beta$, this equation is exactly $W^2W' - \lambda W + u(1 - u) = 0$. Remark 4.8 records the corresponding scaling balance.

1.3. Further related work

The recursions studied here are connected with the broader literature on nonlinear hierarchical systems, in which each generation is obtained by applying a deterministic or random update map to independent copies of the preceding generation. General limit questions for such systems were studied by Shneiberg [16], Li and Rogers [13], and Jordan [12]. For general background on recursive distributional equations and their associated recursive tree processes, see Aldous

and Bandyopadhyay [4]. A distinctive feature of the present model is that its distributional recursion is accompanied by a canonical coupling through the underlying series-parallel networks.

The exact CDF recursion underlying our near-critical analysis admits a continuum interpretation: after rescaling, its leading-order behavior is described by a continuum equation for candidate limiting shapes of the rescaled CDFs. Similar connections arise in hipster random walks and cooperative-motion models [1–3], while Morfe [14] developed this approach for a class of recursive distributional equations that includes the critical resistance recursion. Our proof uses the continuum equation only to identify these candidate shapes and remains otherwise discrete. We use the resulting shapes to construct upper and lower comparison CDFs, verify one-step inequalities directly for the exact recursion, and iterate them using stochastic order. The order preservation and approximation bounds underlying this argument are reminiscent of the Barles–Souganidis framework [6]. The threshold in the associated continuum equation is analogous to minimum-speed thresholds in travelling-wave theory [11, 15].

1.4. Organization and conventions

The remainder of the paper is organized as follows. Section 2 collects the necessary preliminaries. Sections 3 and 4 prove Theorems 1.1–1.2 and Theorem 1.3, respectively. Appendix A and Appendix B establish the ODE analysis used in the upper and lower bound arguments of Section 4.

For real-valued random variables U and V , we write $U \leq_{\text{st}} V$ when U is stochastically dominated by V . We write $x_+ := \max\{x, 0\}$. For positive quantities, $f \asymp g$ means that their ratio is bounded above and below by positive constants in the stated regime. Unless stated otherwise, c and C denote finite positive constants whose values may change from line to line.

AI-assisted formalization methodology

During August 2026, the authors used OpenAI ChatGPT with the GPT-5.6 model identified in the project record and OpenAI Codex with a GPT-5-family model (the client did not expose a more specific deployment identifier). The tools assisted with initial proof sketches, Lean declaration and proof drafts, mathlib search, consistency checks, language editing, and suggestions for the manuscript-formalization crosswalk. They did not serve as authors or as sources of mathematical authority.

The human authors specified the mathematical and graph-semantic contracts, reconstructed and checked the arguments, verified literature claims against primary sources, reviewed the generated Lean source, and accepted or rejected each substantive change. Reproducibility checks use Lean 4.32.1 and mathlib revision 520045ab14e26149ee970e2e617ca04b09bde5d6; they include filewise kernel checking, a clean lake build, proof-escape and placeholder scans, and `#print axioms` fingerprints. The frozen artifact records the specifications, crosswalk, build instructions, audit reports, and two explicit external mathematical interfaces forming the trust boundary. Archive identifiers are given in the Code availability section.

2. Preliminaries

This section collects the preliminary tools used in the proofs of our main results. Section 2.1 gives a unified recursion for distance and resistance and analyzes it on the logarithmic scale, including the one-step change in the mean logarithm. Section 2.2 records some basic structural properties. Section 2.3 specializes to resistance and derives the exact CDF update used in the near-critical analysis.

2.1. Unified recursion and logarithmic update maps

Fix $p \in [0, 1]$. Recall from Section 1.1 that, for this fixed p , the two sequences $(D_n(p))_{n \geq 0}$ and $(R_n(p))_{n \geq 0}$ are jointly constructed from the common environment $(\xi_u)_{u \in \mathbb{T}}$. To treat distance and resistance simultaneously, set

$$Z_n^{(0)}(p) := D_n(p), \quad Z_n^{(1)}(p) := R_n(p).$$

For $\eta \in \{0, 1\}$ and $x, y > 0$, define the corresponding parallel operation by

$$\Pi_\eta(x, y) := \frac{x \wedge y}{\left(1 + \frac{x \wedge y}{x \vee y}\right)^\eta}.$$

Thus $\Pi_0(x, y) = x \wedge y$ and $\Pi_1(x, y) = xy/(x + y)$. Decomposing the environment tree at its root then yields the marginal recursion:

$$Z_{n+1}^{(\eta)} \stackrel{d}{=} \begin{cases} Z_n^{(\eta,1)}(p) + Z_n^{(\eta,2)}(p), & \text{with probability } p, \\ \Pi_\eta\left(Z_n^{(\eta,1)}(p), Z_n^{(\eta,2)}(p)\right), & \text{with probability } 1 - p, \end{cases} \quad (2.1)$$

Here $Z_n^{(\eta,1)}(p)$ and $Z_n^{(\eta,2)}(p)$ are independent copies of $Z_n^{(\eta)}(p)$ and are independent of the choice between series and parallel composition at the root.

We next introduce the series and parallel update maps in logarithmic coordinates. For $x, y \in \mathbb{R}$, define the logarithmic series map and the logarithmic parallel map with parameter η , respectively, by

$$g_+(x, y) = \log(e^x + e^y), \quad g_-^{(\eta)}(x, y) = \begin{cases} x \wedge y, & \eta = 0, \\ -\log(e^{-x} + e^{-y}), & \eta = 1. \end{cases}$$

Thus $g_+(x, y)$ is the logarithm of either the graph distance or the effective resistance obtained by joining two components in series whose corresponding logarithmic quantities are x and y . Likewise, $g_-^{(0)}(x, y)$ and $g_-^{(1)}(x, y)$ are, respectively, the logarithmic graph distance and logarithmic effective resistance obtained by joining such components in parallel.

The definitions immediately give the following properties:

1. The maps are nondecreasing in each coordinate.
2. The maps are translation covariant: for every $x, y, a \in \mathbb{R}$,

$$g_+(x + a, y + a) = a + g_+(x, y), \quad g_-^{(\eta)}(x + a, y + a) = a + g_-^{(\eta)}(x, y). \quad (2.2)$$

We shall use the following max-min representations of these maps. Let

$$h(t) = \log(1 + e^{-t}), \quad t \geq 0.$$

Then

$$g_+(x, y) = (x \vee y) + h(|x - y|), \quad g_-^{(\eta)}(x, y) = (x \wedge y) - \eta h(|x - y|). \quad (2.3)$$

We now set

$$X_n^{(\eta)}(p) := \log Z_n^{(\eta)}(p).$$

Its recursion and mean drift are summarized in the following lemma.

Lemma 2.1. Fix $\eta \in \{0, 1\}$ and $p \in [0, 1]$, and write $X_n := X_n^{(\eta)}(p)$. For $n \geq 0$, if X'_n is an independent copy of X_n , then

$$X_{n+1} \stackrel{d}{=} \begin{cases} g_+(X_n, X'_n), & \text{with probability } p, \\ g_-^{(\eta)}(X_n, X'_n), & \text{with probability } 1 - p, \end{cases} \quad (2.4)$$

where the choice between the two cases is independent of (X_n, X'_n) . Moreover,

$$\mathbb{E}X_{n+1} - \mathbb{E}X_n = \left(p - \frac{1}{2}\right) \mathbb{E}|X_n - X'_n| + \{p - \eta(1 - p)\} \mathbb{E}h(|X_n - X'_n|). \quad (2.5)$$

Proof. The recursion (2.4) is immediate. The identities

$$\mathbb{E}(X_n \vee X'_n) = \mathbb{E}X_n + \frac{1}{2} \mathbb{E}|X_n - X'_n|, \quad \mathbb{E}(X_n \wedge X'_n) = \mathbb{E}X_n - \frac{1}{2} \mathbb{E}|X_n - X'_n|$$

and the logarithmic maps in (2.3) give (2.5). □

2.2. Basic structural properties

Recall the binary-tree representation of the model from Section 1.1. We record several basic structural properties that will be used below. For $n \geq 0$, let

$$\mathcal{F}_n := \sigma(\xi_u : |u| < n).$$

Proposition 2.2 (Conditional refinement property). *For every $n, r \geq 0$, conditionally on $\mathcal{F}_n$, the graph $\mathcal{G}_{n+r}(p)$ is obtained from $\mathcal{G}_n(p)$ by replacing each edge with an independent depth- r copy of $\mathcal{G}_r(p)$.*

Proof. For distinct $u \in \{0, 1\}^n$, the descendant environments $(\xi_{uw})_{w \in \mathbb{T}}$ are independent of one another and of $\mathcal{F}_n$, and each has the same law as the original environment. Restricting these environments to $|w| < r$ gives the claim. $\square$

Lemma 2.3. *For $\eta \in \{0, 1\}$, $p \in [0, 1]$, and $n \geq 0$, we have*

$$2^{-\eta} Z_n^{(\eta)}(p) \leq Z_{n+1}^{(\eta)}(p) \leq 2 Z_n^{(\eta)}(p) \quad \text{almost surely.} \quad (2.6)$$

Consequently,

$$-\eta n \log 2 \leq X_n^{(\eta)}(p) \leq n \log 2 \quad \text{almost surely,} \quad \left| \mathbb{E} X_{n+1}^{(\eta)}(p) - \mathbb{E} X_n^{(\eta)}(p) \right| \leq \log 2. \quad (2.7)$$

Proof. For distance, each old edge is replaced by a two-edge network whose boundary-to-boundary distance is either 1 or 2. Thus $D_n(p) \leq D_{n+1}(p) \leq 2D_n(p)$. For resistance, every old unit edge is replaced by a two-edge network of effective resistance $1/2$ or 2 . Rayleigh monotonicity and homogeneity compare the new network with the old one after all edge resistances are multiplied by $1/2$ or by 2 . Together, these comparisons prove (2.6), which immediately yields (2.7). $\square$

Lemma 2.4. *For $\eta \in \{0, 1\}$, $p \in [0, 1]$, and $n \geq 0$,*

$$\mathbb{E} Z_{n+1}^{(\eta)}(p) \geq 2p \mathbb{E} Z_n^{(\eta)}(p), \quad (2p)^n \leq \mathbb{E} Z_n^{(\eta)}(p) \leq 2^n. \quad (2.8)$$

Proof. Taking expectations in (2.1) gives the first inequality. Its iteration, together with $Z_0^{(\eta)}(p) = 1$ and Lemma 2.3, gives the remaining bounds. $\square$

We next record resistance duality. For an environment $\xi = (\xi_u)_{u \in \mathbb{T}}$, write $R_n(\xi)$ for the resistance of the resulting depth- n network and put $\bar{\xi} = (1 - \xi_u)_{u \in \mathbb{T}}$. If ξ has parameter p , then $\bar{\xi}$ has parameter $1 - p$; we use these two environments to couple $R_n(p)$ and $R_n(1 - p)$. Complementing the environment exchanges series and parallel at every internal node of the construction tree. Since $g_-^{(1)}(x, y) = -g_+(-x, -y)$, induction using (2.1) gives the following.

Lemma 2.5. *Under this coupling, for every $p \in [0, 1]$ and $n \geq 0$,*

$$\log R_n(p) = -\log R_n(1 - p) \quad \text{almost surely.} \quad (2.9)$$

2.3. Exact resistance CDF operator

In this subsection, we consider only the resistance model and write $g_- := g_-^{(1)}$ and $X_n(p) := X_n^{(1)}(p)$.

Let U and V be independent random variables with common CDF F . Denote the CDFs of $g_+(U, V)$ and $g_-(U, V)$ by S_F and P_F , respectively:

$$S_F(x) := \mathbb{P}(g_+(U, V) \leq x), \quad P_F(x) := \mathbb{P}(g_-(U, V) \leq x).$$

For $p \in [0, 1]$, let A be independent of (U, V) , take values in $\{+, -\}$, and satisfy $\mathbb{P}(A = +) = p$ and $\mathbb{P}(A = -) = 1 - p$. Define

$$(\mathcal{T}_p F)(x) := \mathbb{P}(g_A(U, V) \leq x) = p S_F(x) + (1 - p) P_F(x). \quad (2.10)$$

Thus $\mathcal{T}_p$ is the one-step CDF update for the resistance recursion. In particular, if F is the CDF of $X_n(p)$, then $\mathcal{T}_p F$ is the CDF of $X_{n+1}(p)$.

We now give an explicit formula for this update when F is absolutely continuous. Let ρ be the density of F , and recall that $h(r) = \log(1 + e^{-r})$ for $r \geq 0$. Define

$$I_-(F; x) := \int_0^\infty \int_0^{h(r)} \rho(x-s)\rho(x-s-r) \, ds \, dr,$$

$$I_+(F; x) := \int_0^\infty \int_0^{h(r)} \rho(x+s)\rho(x+s+r) \, ds \, dr.$$

Lemma 2.6. *If F is absolutely continuous with density ρ , then S_F and P_F satisfy*

$$S_F(x) = F(x)^2 - 2I_-(F; x), \quad P_F(x) = 2F(x) - F(x)^2 + 2I_+(F; x). \quad (2.11)$$

Consequently, if $p = 1/2 + \delta$, then

$$\mathcal{T}_{1/2+\delta}F - F = I_+ - I_- - 2\delta F(1 - F) - 2\delta(I_+ + I_-). \quad (2.12)$$

Moreover, the operator $\mathcal{T}_p$ preserves pointwise order and commutes with translations. More precisely, for every $p \in [0, 1]$ and any CDFs F and G , the following properties hold:

(i) Order preservation.

$$F \leq G \text{ pointwise} \implies \mathcal{T}_p F \leq \mathcal{T}_p G \text{ pointwise.}$$

(ii) Translation covariance. For $a \in \mathbb{R}$, define $(\tau_a F)(x) := F(x - a)$. Then

$$\mathcal{T}_p(\tau_a F) = \tau_a(\mathcal{T}_p F).$$

Proof. By (2.3),

$$g_+(U, V) = (U \vee V) + h(|U - V|).$$

In particular, $g_+(U, V) \geq U \vee V$. Since U and V have densities, the diagonal has probability zero, and symmetry between U and V gives

$$S_F(x) = F(x)^2 - 2\mathbb{P}(V \leq U \leq x, U + h(U - V) > x).$$

On the event inside the last probability, make the change of variables

$$r = U - V \geq 0, \quad s = x - U \geq 0.$$

Its Jacobian has absolute value one, and the crossing condition is precisely $s < h(r)$. Therefore

$$\begin{aligned} \mathbb{P}(V \leq U \leq x, U + h(U - V) > x) &= \int_0^\infty \int_0^{h(r)} \rho(x-s)\rho(x-s-r) \, ds \, dr \\ &= I_-(F; x). \end{aligned}$$

This proves the formula for S_F .

The formula for P_F follows by the same argument, using (2.3). This proves (2.11).

Substituting these two formulas into (2.10), with $p = 1/2 + \delta$, yields

$$\begin{aligned} \mathcal{T}_{1/2+\delta}F - F &= \left(\frac{1}{2} + \delta\right)(F^2 - 2I_-) + \left(\frac{1}{2} - \delta\right)(2F - F^2 + 2I_+) - F \\ &= I_+ - I_- - 2\delta F(1 - F) - 2\delta(I_+ + I_-). \end{aligned}$$

This is (2.12).

It remains to verify the two structural properties of the operator. Suppose that $F \leq G$ pointwise. By the standard monotone coupling, there exist $U_F, V_F \stackrel{\text{i.i.d.}}{\sim} F$ and $U_G, V_G \stackrel{\text{i.i.d.}}{\sim} G$ on a common probability space such that $U_F \geq U_G$ and $V_F \geq V_G$ almost surely. The coordinatewise monotonicity of g_+ and g_- gives

$$S_F \leq S_G, \quad P_F \leq P_G.$$

Hence $\mathcal{T}_p F \leq \mathcal{T}_p G$, proving preservation of pointwise order. Finally, $\tau_a F$ is the CDF of $U + a$. Translation covariance (2.2) gives

$$(\mathcal{T}_p(\tau_a F))(x) = (\mathcal{T}_p F)(x - a) = (\tau_a(\mathcal{T}_p F))(x),$$

so $\mathcal{T}_p$ commutes with translations. □

3. Proof of Theorems 1.1 and 1.2

In this section, we prove Theorems 1.1 and 1.2. Section 3.1 establishes the existence of the first-moment logarithmic rates asserted in Theorem 1.2 via first-moment submultiplicativity; see Lemma 3.1. Section 3.2 proves the assertions of both theorems in the supercritical regime $p > 1/2$ by controlling the Jensen gap defined in (3.3). Section 3.3 completes the proofs by treating $p \leq 1/2$.

Throughout this section, fix $\eta \in \{0, 1\}$ and $p \in [0, 1]$. For notational convenience, we suppress these parameters whenever no confusion can arise and write

$$Z_n := Z_n^{(\eta)}(p), \quad X_n := X_n^{(\eta)}(p).$$

3.1. First-moment logarithmic rate

Lemma 3.1. *For each $\eta \in \{0, 1\}$ and all $n, r \geq 0$,*

$$\mathbb{E}Z_{n+r} \leq \mathbb{E}Z_n \mathbb{E}Z_r. \quad (3.1)$$

Consequently, the limit

$$\gamma = \gamma^{(\eta)}(p) := \lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}Z_n = \inf_{n \geq 1} \frac{1}{n} \log \mathbb{E}Z_n \quad (3.2)$$

exists in $(-\infty, \infty)$.

For distance, Chen, Derrida, Duquesne, and Shi [7, the argument leading to equation (1.2)] established (3.1) and (3.2). We reproduce the proof for completeness.

Proof. Recall the conditional refinement property in Proposition 2.2. Conditionally on $\mathcal{F}_n$, each edge of $\mathcal{G}_n$ is replaced in $\mathcal{G}_{n+r}$ by an independent depth- r copy of $\mathcal{G}_r$.

For distance, fix a shortest path in $\mathcal{G}_n$, using a deterministic tie-breaking rule. In each depth- r copy replacing an edge of this path, choose a shortest path between its boundary vertices. Concatenating these paths gives a path across $\mathcal{G}_{n+r}$. Conditional on $\mathcal{F}_n$, its expected length is $(\mathbb{E}D_r)D_n$, and hence

$$\mathbb{E}[D_{n+r} \mid \mathcal{F}_n] \leq (\mathbb{E}D_r)D_n.$$

Taking expectations proves (3.1) for distance.

For resistance, we use Thomson's principle in its unit-resistance form:

$$R_{\text{eff}}(G) = \inf_{\theta} \sum_{e \in E(G)} \theta_e^2,$$

where the infimum is over unit flows between the boundary vertices of G . Let (θ_e) be the minimizing unit flow in $\mathcal{G}_n$. In the depth- r copy replacing e , take a minimizing unit flow and multiply it by θ_e . Combining these flows gives a unit flow in $\mathcal{G}_{n+r}$ with energy $\sum_e \theta_e^2 R_r^{(e)}$. Therefore, $R_{n+r} \leq \sum_e \theta_e^2 R_r^{(e)}$. Taking conditional expectation gives

$$\mathbb{E}[R_{n+r} \mid \mathcal{F}_n] \leq (\mathbb{E}R_r) \sum_e \theta_e^2 = (\mathbb{E}R_r)R_n.$$

Taking expectations proves (3.1) for resistance.

Fekete's subadditive lemma, together with Lemma 2.3, proves (3.2). □

3.2. From first-moment logarithmic rate to logarithmic speed

Throughout this subsection, we assume that $p > 1/2$. Define the Jensen gap by

$$J_n := \log \mathbb{E}Z_n - \mathbb{E}X_n \geq 0. \quad (3.3)$$

We shall first prove that J_n is uniformly bounded, showing that $\mathbb{E}X_n$ and $\log \mathbb{E}Z_n$ have the same limiting rate after division by n , and then use Lemma 3.3 to control the centered logarithmic fluctuations. Together, these estimates yield the L^1 convergence in (1.1) and the almost-sure convergence.

Proposition 3.2. *For $p > 1/2$, the Jensen gaps are uniformly bounded:*

$$\sup_{n \geq 0} J_n < \infty.$$

Consequently,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} X_n = \lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} Z_n = \gamma. \quad (3.4)$$

We defer the proof of Proposition 3.2. Assuming it, we next prove Theorem 1.1 for $p > 1/2$. We begin with a preliminary lemma.

Lemma 3.3. *For $p > 1/2$, let X'_n be an independent copy of X_n . With $C := \log 2/(p - 1/2)$, we have*

$$\sup_{n \geq 0} \mathbb{E} |X_n - X'_n| \leq C, \quad \sup_{n \geq 0} \mathbb{E} |X_n - \mathbb{E} X_n| \leq C. \quad (3.5)$$

Consequently,

$$\frac{X_n - \mathbb{E} X_n}{n} \longrightarrow 0 \quad \text{a.s. and in } L^1. \quad (3.6)$$

Proof. Since $p > 1/2$, the coefficient of the h term in the drift identity (2.5) is nonnegative: it equals p for distance and $2p - 1$ for resistance. Combining that identity with the adjacent-generation bound in Lemma 2.3, we obtain

$$0 \leq \left(p - \frac{1}{2}\right) \mathbb{E} |X_n - X'_n| \leq \mathbb{E} X_{n+1} - \mathbb{E} X_n \leq \log 2,$$

which proves the first bound in (3.5). Conditional Jensen gives

$$\mathbb{E} |X_n - \mathbb{E} X_n| = \mathbb{E} |X_n - \mathbb{E}[X'_n | X_n]| \leq \mathbb{E} |X_n - X'_n|,$$

which proves the second bound. The L^1 convergence in (3.6) follows immediately.

For the almost-sure statement, Markov's inequality and (3.5) give, for every $j \geq 1$,

$$\sum_{k \geq 1} \mathbb{P} \left\{ |X_{k^2} - \mathbb{E} X_{k^2}| > \frac{k^2}{j} \right\} \leq jC \sum_{k \geq 1} \frac{1}{k^2} < \infty.$$

Thus Borel–Cantelli yields

$$\frac{X_{k^2} - \mathbb{E} X_{k^2}}{k^2} \longrightarrow 0 \quad \text{almost surely.}$$

If $k^2 \leq n \leq (k+1)^2$, Lemma 2.3 gives

$$\left| (X_n - \mathbb{E} X_n) - (X_{k^2} - \mathbb{E} X_{k^2}) \right| \leq 2(2k+1) \log 2.$$

Dividing by n completes the proof. $\square$

Proof of Theorems 1.1 and 1.2 for $p > 1/2$. Section 3.1 establishes the existence of the first-moment logarithmic rates $\gamma^{(0)}(p) = \gamma_D(p)$ and $\gamma^{(1)}(p) = \gamma_R(p)$. Proposition 3.2 and Lemma 3.3 then imply

$$\frac{X_n}{n} \longrightarrow \gamma = \gamma^{(\eta)}(p) \quad \text{a.s. and in } L^1. \quad (3.7)$$

Setting $\eta = 0$ and $\eta = 1$ in (3.7) proves that $\gamma_D(p) = v_D(p)$ and $\gamma_R(p) = v_R(p)$, respectively, for $p > 1/2$, which proves (1.4) in this range. Finally, (2.8) and (3.2) prove (1.2). $\square$

We now prove Proposition 3.2. The idea is as follows. We first use the second-moment method to show that $Z_n \geq \mathbb{E}Z_n/2$ with uniformly positive probability. On this event, we have $J_n \leq X_n - \mathbb{E}X_n + \log 2$; since its probability is bounded away from zero, the second bound in (3.5) yields the desired uniform bound on J_n .

We begin by bounding the second-moment ratio. Set

$$K_n := \frac{\mathbb{E}Z_n^2}{(\mathbb{E}Z_n)^2}.$$

Lemma 3.4. *For $p > 1/2$,*

$$\sup_{n \geq 0} K_n \leq K := \frac{2p + (1-p)4^{-\eta}}{2p(2p-1)}.$$

Proof. Let Z'_n be an independent copy of Z_n . We have

$$\Pi_0(Z_n, Z'_n)^2 \leq Z_n Z'_n, \quad \Pi_1(Z_n, Z'_n)^2 \leq \frac{Z_n Z'_n}{4}.$$

Using independence and $\mathbb{E}(Z_n + Z'_n)^2 = 2\mathbb{E}Z_n^2 + 2(\mathbb{E}Z_n)^2$, we obtain

$$\mathbb{E}Z_{n+1}^2 = p\mathbb{E}(Z_n + Z'_n)^2 + (1-p)\mathbb{E}\Pi_\eta(Z_n, Z'_n)^2 \leq 2p\mathbb{E}Z_n^2 + \{2p + (1-p)4^{-\eta}\}(\mathbb{E}Z_n)^2.$$

After division by $(\mathbb{E}Z_{n+1})^2$ and using (2.8),

$$K_{n+1} \leq \frac{1}{2p}K_n + \frac{2p + (1-p)4^{-\eta}}{4p^2}. \quad (3.8)$$

The affine map on the right-hand side of (3.8) has slope $1/(2p) < 1$ and fixed point K . Since $K_0 = 1 \leq K$, induction gives $K_n \leq K$ for every $n \geq 0$. $\square$

Proof of Proposition 3.2. By Lemma 3.4 and the Paley–Zygmund inequality at threshold $1/2$,

$$\mathbb{P}\left\{Z_n \geq \frac{\mathbb{E}Z_n}{2}\right\} \geq \frac{1}{4K}. \quad (3.9)$$

Observe that on the event in (3.9),

$$X_n - \mathbb{E}X_n \geq \log(\mathbb{E}Z_n/2) - \mathbb{E}X_n = J_n - \log 2. \quad (3.10)$$

Consequently, by (3.9),

$$\frac{(J_n - \log 2)_+}{4K} \leq \mathbb{E}\left[(J_n - \log 2)_+ \mathbf{1}_{\{Z_n \geq \mathbb{E}Z_n/2\}}\right] \leq \mathbb{E}\left[|X_n - \mathbb{E}X_n| \mathbf{1}_{\{Z_n \geq \mathbb{E}Z_n/2\}}\right] \leq \mathbb{E}|X_n - \mathbb{E}X_n| \leq C.$$

Therefore,

$$0 \leq J_n \leq \log 2 + 4KC. \quad (3.11)$$

Combining Lemma 3.1 with (3.11),

$$\frac{\mathbb{E}X_n}{n} = \frac{\log \mathbb{E}Z_n}{n} - \frac{J_n}{n} \longrightarrow \gamma,$$

which proves (3.4) and completes the proof. $\square$

3.3. Remaining parameter ranges

Proof of Theorem 1.1 for $p \leq 1/2$. For distance, Chen, Derrida, Duquesne, and Shi [7] showed that, for $p \leq 1/2$, its first-moment logarithmic rate vanishes, that is,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} D_n(p) = 0; \quad (3.12)$$

Since $D_n(p) \geq 1$, Jensen's inequality gives

$$0 \leq \frac{1}{n} \mathbb{E} \log D_n(p) \leq \frac{1}{n} \log \mathbb{E} D_n(p) \longrightarrow 0.$$

Thus $n^{-1} \log D_n(p) \rightarrow 0$ in L^1 . The convergence also holds almost surely. Indeed, for every $\varepsilon > 0$, (3.12) gives, for all sufficiently large n ,

$$\mathbb{P}\{\log D_n(p) > \varepsilon n\} \leq e^{-\varepsilon n} \mathbb{E} D_n(p) \leq e^{-\varepsilon n/2}.$$

Borel–Cantelli and $D_n(p) \geq 1$ prove that $n^{-1} \log D_n(p) \rightarrow 0$ almost surely.

For $p < 1/2$, define $v_R(p) = -v_R(1-p)$. Since $1-p > 1/2$, (3.7) at parameter $1-p$, together with Lemma 2.5, gives

$$\frac{1}{n} \log R_n(p) \longrightarrow -v_R(1-p) = v_R(p) \quad \text{a.s. and in } L^1.$$

At $p = 1/2$, let ξ be the parameter-1/2 environment and let $\bar{\xi}$ be its complement. Thomson's principle applied to a unit flow along a shortest path, together with (2.9), gives

$$-\log D_n(\bar{\xi}) \leq \log R_n(\xi) \leq \log D_n(\xi).$$

Since $D_n(\xi), D_n(\bar{\xi}) \geq 1$, the preceding sandwich implies

$$|\log R_n(\xi)| \leq \log D_n(\xi) + \log D_n(\bar{\xi}).$$

The L^1 and almost-sure distance result proved above applies to both ξ and $\bar{\xi}$. Dividing by n and taking expectations proves the L^1 convergence; the almost-sure convergence follows from the same inequality. $\square$

Proof of Theorem 1.2 for $p \leq 1/2$. For distance, (3.12) and Theorem 1.1 give $\gamma_D(p) = v_D(p) = 0$ for $p \leq 1/2$.

For resistance, $R_n(0) = 2^{-n}$, so $\gamma_R(0) = v_R(0) = -\log 2$. At $p = 1/2$, Jensen's inequality, the L^1 convergence established above, the bound $R_n(1/2) \leq D_n(1/2)$, and (3.12) yield

$$0 = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \log R_n(1/2) \leq \liminf_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} R_n(1/2) \leq \limsup_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} R_n(1/2) \leq \lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E} D_n(1/2) = 0.$$

Thus $\gamma_R(1/2) = v_R(1/2) = 0$, which proves (1.4) at $p = 1/2$.

For $0 < p < 1/2$, Jensen's inequality and the L^1 convergence in Theorem 1.1 give $\gamma_R(p) \geq v_R(p)$. If $\gamma_R(p) = v_R(p)$, then Lemma 2.4 gives $\gamma_R(p) \geq \log(2p)$, and the desired identity follows. It therefore suffices to prove that if $\gamma_R(p) > v_R(p)$, then $\gamma_R(p) = \log(2p)$.

Suppose then that $\gamma_R(p) > v_R(p)$ and set $m_n = \mathbb{E} R_n(p)$ and

$$\hat{R}_n = \frac{R_n(p)}{m_n}.$$

Then $\mathbb{E} \hat{R}_n = 1$ and

$$\frac{1}{n} \log \hat{R}_n = \frac{1}{n} \log R_n(p) - \frac{1}{n} \log m_n \longrightarrow v_R(p) - \gamma_R(p) < 0 \quad \text{a.s.}$$

Hence $\hat{R}_n \rightarrow 0$ a.s. If $\hat{R}'_n$ is an independent copy of $\hat{R}_n$, then, for every $\tau > 0$,

$$c_n := \mathbb{E} \frac{\hat{R}_n \hat{R}'_n}{\hat{R}_n + \hat{R}'_n} \leq \mathbb{E}(\hat{R}_n \wedge \hat{R}'_n) \leq \tau + \mathbb{E}[\hat{R}_n \mathbf{1}_{\{\hat{R}'_n > \tau\}}] = \tau + \mathbb{P}(\hat{R}_n > \tau).$$

Consequently $c_n \rightarrow 0$, and the exact first-moment recursion yields

$$\frac{m_{n+1}}{m_n} = 2p + (1-p)c_n \longrightarrow 2p,$$

$$\gamma_R(p) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} \log \frac{m_{j+1}}{m_j} = \log(2p).$$

This completes the proof. $\square$

4. Proof of Theorem 1.3

Section 4 first collects the constants and functions used in the proof and then proves the upper and lower bounds. From this section onward, we specialize to effective resistance and write $v(p) := v_R(p)$.

4.1. Preliminary

The following lemma calculates the coefficient appearing in the later estimates in Lemmas 4.4 and 4.6. Recall that $h(r) = \log(1 + e^{-r})$ for $r \geq 0$.

Lemma 4.1. *The following equality holds:*

$$\int_0^\infty (h(r)^2 + rh(r)) \, dr = \zeta(3).$$

Proof. With $u = (1 + e^r)^{-1}$, symmetry under $u \mapsto 1 - u$ gives

$$\begin{aligned} \int_0^\infty (h(r)^2 + rh(r)) \, dr &= \int_0^{1/2} \frac{\log u \log(1-u)}{u(1-u)} \, du \\ &= \int_0^1 \frac{\log u \log(1-u)}{u} \, du \\ &= \sum_{n \geq 1} \frac{1}{n} \int_0^1 u^{n-1} (-\log u) \, du = \sum_{n \geq 1} \frac{1}{n^3} = \zeta(3), \end{aligned}$$

where the first line also uses $1/[u(1-u)] = 1/u + 1/(1-u)$, and the second follows from $-\log(1-u) = \sum_{n \geq 1} u^n/n$ and Tonelli's theorem. $\square$

For later use set

$$a = 2 \int_0^\infty (h(r)^2 + rh(r)) \, dr = 2\zeta(3), \quad \beta = \left(\frac{2}{a}\right)^{1/3}, \quad \kappa_\lambda = \frac{2\lambda}{\beta} = 2\zeta(3)^{1/3}\lambda. \quad (4.1)$$

The following two propositions, proved in Appendix B, provide the distribution functions and their properties needed for the upper and lower bounds, respectively. Let W_λ denote the function constructed in Appendix A from the solution y_λ of (A.5).

Proposition 4.2. *For every $\lambda > \lambda_*$, there is a unique increasing $\Phi_\lambda \in C^\infty(\mathbb{R}; (0, 1))$ such that*

$$\begin{aligned} \Phi'_\lambda(z) &= \beta W_\lambda(1 - \Phi_\lambda(z)), \\ \Phi_\lambda(0) &= \frac{1}{2}, \quad \lim_{z \rightarrow -\infty} \Phi_\lambda(z) = 0, \quad \lim_{z \rightarrow \infty} \Phi_\lambda(z) = 1. \end{aligned} \quad (4.2)$$

Put $q_\lambda = \Phi'_\lambda$. Then $q_\lambda > 0$ and

$$aq_\lambda q'_\lambda - 2\Phi_\lambda(1 - \Phi_\lambda) = -\kappa_\lambda q_\lambda \quad \text{on } \mathbb{R}, \quad (4.3)$$

and there exists $M_\lambda < \infty$ such that

$$\max_{1 \leq j \leq 3} |q_\lambda^{(j)}(z)| \leq M_\lambda q_\lambda(z), \quad z \in \mathbb{R}. \quad (4.4)$$

Moreover,

$$\frac{q_\lambda(z)}{\Phi_\lambda(z)} \longrightarrow \frac{\beta}{\lambda} \quad (z \rightarrow -\infty), \quad \frac{q_\lambda(z)}{1 - \Phi_\lambda(z)} \longrightarrow \frac{\beta}{\lambda} \quad (z \rightarrow \infty). \quad (4.5)$$

Consequently, there exists $c_\lambda, C_\lambda > 0$ such that

$$\Phi_\lambda(z) \leq C_\lambda e^{c_\lambda z} \quad (z \leq 0), \quad 1 - \Phi_\lambda(z) \leq C_\lambda e^{-c_\lambda z} \quad (z \geq 0), \quad \int_{\mathbb{R}} |x| \, d\Phi_\lambda(x) < \infty.$$

Proposition 4.3. For every $0 < \lambda < \lambda_*$, there is a unique increasing $\Psi_\lambda \in C^\infty([0, \infty); [0, 1])$ such that

$$\begin{aligned} \Psi'_\lambda(z) &= \beta W_\lambda(\Psi_\lambda(z)), \\ \Psi_\lambda(0) &= 0, \quad \Psi_\lambda(z) \longrightarrow 1 \quad (z \rightarrow \infty). \end{aligned} \quad (4.6)$$

When extended by zero to $(-\infty, 0)$, Ψ_λ is an absolutely continuous CDF supported on $[0, \infty)$.

Put $q_\lambda = \Psi'_\lambda$ on $[0, \infty)$. At the endpoint 0, all derivatives are taken from the right. Then

$$q_\lambda(0) = \beta W_\lambda(0) > 0, \quad a q_\lambda q'_\lambda + 2\Psi_\lambda(1 - \Psi_\lambda) = \kappa_\lambda q_\lambda \quad (z \geq 0), \quad (4.7)$$

and there exists $M_\lambda < \infty$ such that

$$\max_{1 \leq j \leq 3} |q_\lambda^{(j)}(z)| \leq M_\lambda q_\lambda(z), \quad z \geq 0. \quad (4.8)$$

Furthermore, q_λ has a strictly positive extension $\bar{q}_\lambda \in C^3(\mathbb{R})$, equal to q_λ on $[0, \infty)$, such that, for some $\bar{M}_\lambda < \infty$,

$$\max_{1 \leq j \leq 3} |\bar{q}_\lambda^{(j)}(z)| \leq \bar{M}_\lambda \bar{q}_\lambda(z), \quad z \in \mathbb{R}. \quad (4.9)$$

Moreover,

$$\frac{q_\lambda(z)}{1 - \Psi_\lambda(z)} \longrightarrow \frac{\beta}{\lambda} \quad (z \rightarrow \infty). \quad (4.10)$$

Consequently, there exists $c_\lambda, C_\lambda > 0$ such that

$$1 - \Psi_\lambda(z) \leq C_\lambda e^{-c_\lambda z} \quad (z \geq 0), \quad \int_{[0, \infty)} x \, d\Psi_\lambda(x) < \infty.$$

Throughout this section $\varepsilon > 0$ is small and

$$p_+ = \frac{1}{2} + \varepsilon^3, \quad p_- = \frac{1}{2} - \varepsilon^3.$$

Here and below, the bias means $p - \frac{1}{2}$; thus p_+ has positive bias and p_- has negative bias. Let $F_n^\pm$ denote the CDFs of $X_n(p_\pm)$, respectively.

4.2. Upper bound

Fix $\lambda > \lambda_*$ and $\kappa > \kappa_\lambda$. It is enough to construct an integrable Y_ε with CDF G_ε satisfying

$$F_0 \geq G_\varepsilon, \quad (4.11)$$

$$\mathcal{T}_{p_+} G_\varepsilon(x) \geq G_\varepsilon(x - \kappa \varepsilon^2), \quad x \in \mathbb{R}, \quad (4.12)$$

where F_0 is the CDF of $X_0 = 0$.

Indeed, if such G_ε exists, then order preservation and translation covariance in Lemma 2.6 give inductively

$$F_n^+(x) \geq G_\varepsilon(x - n\kappa\varepsilon^2).$$

We prove this by induction: If the inequality holds at time n , then

$$F_{n+1}^+ = \mathcal{T}_{p_+} F_n^+ \geq \mathcal{T}_{p_+} (\tau_{n\kappa\varepsilon^2} G_\varepsilon) = \tau_{n\kappa\varepsilon^2} (\mathcal{T}_{p_+} G_\varepsilon) \geq \tau_{(n+1)\kappa\varepsilon^2} G_\varepsilon.$$

Consequently,

$$X_n(p_+) \leq_{\text{st}} n\kappa\varepsilon^2 + Y_\varepsilon.$$

Since $X_n(p_+)$ and Y_ε are integrable, taking expectations, dividing by n , and using Theorem 1.1 gives

$$\frac{\mathbb{E}X_n(p_+)}{n} \leq \kappa\varepsilon^2 + \frac{\mathbb{E}Y_\varepsilon}{n} \longrightarrow \kappa\varepsilon^2.$$

Thus

$$v(p_+) \leq \kappa\varepsilon^2. \quad (4.13)$$

Taking $\delta = \varepsilon^3$ in (4.13) yields

$$\limsup_{\delta \downarrow 0} \frac{v(\frac{1}{2} + \delta)}{\delta^{2/3}} \leq \kappa.$$

Letting first $\kappa \downarrow \kappa_\lambda$ and then $\lambda \downarrow \lambda_*$, and using (4.1), gives the upper bound

$$\limsup_{\delta \downarrow 0} \frac{v(\frac{1}{2} + \delta)}{\delta^{2/3}} \leq 2\zeta(3)^{1/3} \lambda_*. \quad (4.14)$$

It remains to construct an integrable Y_ε whose CDF G_ε has the two properties in (4.11) and (4.12).

Step 1: Construct a candidate satisfying (4.12) Let $\Phi = \Phi_\lambda$ and $q = \Phi'$ be as in Proposition 4.2. For $z_0 \in \mathbb{R}$ and $\varepsilon > 0$, set

$$F_{\varepsilon, z_0}(x) = \Phi(z_0 + \varepsilon x), \quad \xi = z_0 + \varepsilon x.$$

The following lemma gives the required estimates for Φ .

Lemma 4.4. *There are constants $C_\lambda < \infty$ and $\varepsilon_\lambda > 0$ such that, for all $x, z_0 \in \mathbb{R}$ and $0 < \varepsilon < \varepsilon_\lambda$,*

$$\left| I_+(F_{\varepsilon, z_0}; x) - I_-(F_{\varepsilon, z_0}; x) - a\varepsilon^3 q(\xi)q'(\xi) \right| \leq C_\lambda \varepsilon^5 q(\xi), \quad (4.15)$$

$$I_+(F_{\varepsilon, z_0}; x) + I_-(F_{\varepsilon, z_0}; x) \leq C_\lambda \varepsilon^2 q(\xi). \quad (4.16)$$

Combining Lemmas 2.6 and 4.4 with (4.3) gives

$$\mathcal{T}_{p_+} F_{\varepsilon, z_0}(x) - F_{\varepsilon, z_0}(x) = -\kappa_\lambda \varepsilon^3 q(\xi) + O_\lambda(\varepsilon^5 q(\xi)),$$

for all $x, z_0 \in \mathbb{R}$. Taylor expansion gives

$$F_{\varepsilon, z_0}(x - \kappa\varepsilon^2) - F_{\varepsilon, z_0}(x) = -\kappa\varepsilon^3 q(\xi) + O_{\lambda, \kappa}(\varepsilon^6 q(\xi)).$$

Hence, for all sufficiently small ε and all $x, z_0 \in \mathbb{R}$,

$$\mathcal{T}_{p_+} F_{\varepsilon, z_0}(x) \geq F_{\varepsilon, z_0}(x - \kappa\varepsilon^2) + \frac{\kappa - \kappa_\lambda}{2} \varepsilon^3 q(\xi). \quad (4.17)$$

Thus F_{ε, z_0} satisfies the required inequality (4.12), with a strict margin. However, it does not satisfy (4.11), since $F_{\varepsilon, z_0}(x) > 0 = F_0(x)$ for $x < 0$.

Step 2: Use truncation to construct G_ε Set

$$\alpha_\varepsilon = \varepsilon^4, \quad \Phi(z_\varepsilon) = \alpha_\varepsilon, \quad F_\varepsilon(x) = \Phi(z_\varepsilon + \varepsilon x).$$

Let $\tilde{Y}_\varepsilon$ have CDF F_ε , put $Y_\varepsilon = \tilde{Y}_\varepsilon \vee 0$, and denote its CDF by

$$G_\varepsilon(x) = \begin{cases} 0, & x < 0, \\ F_\varepsilon(x), & x \geq 0, \end{cases} \quad G_\varepsilon(0) = \alpha_\varepsilon.$$

Then $F_0 \geq G_\varepsilon$. The truncation error is controlled by the following lemma.

Lemma 4.5. *For all sufficiently small ε and all $x \geq \kappa\varepsilon^2$,*

$$0 \leq \mathcal{T}_{p_+} F_\varepsilon(x) - \mathcal{T}_{p_+} G_\varepsilon(x) \leq C_\lambda \varepsilon^4 q(z_\varepsilon + \varepsilon x). \quad (4.18)$$

For all sufficiently small ε and $x \geq \kappa\varepsilon^2$, Lemma 4.5 and (4.17) give

$$\begin{aligned} \mathcal{T}_{p_+} G_\varepsilon(x) &\geq F_\varepsilon(x - \kappa\varepsilon^2) + \left(\frac{\kappa - \kappa_\lambda}{2} \varepsilon^3 - C_\lambda \varepsilon^4 \right) q(z_\varepsilon + \varepsilon x) \\ &\geq G_\varepsilon(x - \kappa\varepsilon^2). \end{aligned}$$

Here the last inequality uses the definition of G_ε : $F_\varepsilon(x - \kappa\varepsilon^2) = G_\varepsilon(x - \kappa\varepsilon^2)$ for $x - \kappa\varepsilon^2 \geq 0$.

For $x < \kappa\varepsilon^2$, $G_\varepsilon(x - \kappa\varepsilon^2) = 0$, while $\mathcal{T}_{p_+} G_\varepsilon(x) \geq 0$ because it is a CDF. Thus the two cases give the global inequality

$$\mathcal{T}_{p_+} G_\varepsilon(x) \geq G_\varepsilon(x - \kappa\varepsilon^2), \quad x \in \mathbb{R}.$$

Moreover, Proposition 4.2 implies that the law with CDF Φ has a finite absolute first moment. Hence $\tilde{Y}_\varepsilon$, and therefore Y_ε , is integrable. Thus G_ε has the two properties used above.

We now prove the two estimates used in the construction.

Proof of Lemma 4.4. The density of F_{ε, z_0} is $\rho(x) = \varepsilon q(z_0 + \varepsilon x)$. For fixed $r \geq 0$ and $0 \leq s \leq h(r)$, set

$$Q_{s,r}(t) = q(\xi + ts)q(\xi + t(s+r)).$$

Then

$$I_+(F_{\varepsilon, z_0}; x) - I_-(F_{\varepsilon, z_0}; x) = \varepsilon^2 \int_0^\infty \int_0^{h(r)} (Q_{s,r}(\varepsilon) - Q_{s,r}(-\varepsilon)) \, ds \, dr.$$

The Taylor expansion gives

$$Q_{s,r}(\varepsilon) - Q_{s,r}(-\varepsilon) = 2\varepsilon(2s+r)q(\xi)q'(\xi) + R_{s,r},$$

where

$$|R_{s,r}| \leq \frac{\varepsilon^3}{3} \sup_{|t| \leq \varepsilon} |Q_{s,r}'''(t)|.$$

To bound the remainder, we first derive two estimates on q . Write $M := M_\lambda$, and let C_λ denote constants that may change from line to line. Since $q > 0$, the case $j = 1$ in (4.4) gives

$$e^{-M|y|}q(z) \leq q(z+y) \leq e^{M|y|}q(z), \quad z, y \in \mathbb{R}. \quad (4.19)$$

The monotonicity and range of Φ imply $\int_{\mathbb{R}} q \leq 1$. Hence

$$1 \geq \int_{z-1}^{z+1} q(u) \, du \geq 2e^{-M}q(z), \quad \sup_{z \in \mathbb{R}} q(z) \leq \frac{e^M}{2}. \quad (4.20)$$

Thus every occurrence of $q(z)^2$ below is bounded by a constant times $q(z)$, uniformly in z .

By Leibniz's rule and (4.4), each term in $Q'''_{s,r}(t)$ is bounded, up to a constant depending only on M , by $s^j(s+r)^{3-j}q(\xi+ts)q(\xi+t(s+r))$ for some $0 \leq j \leq 3$. Since $0 \leq s \leq h(r) \leq \log 2$, (4.19) therefore gives

$$\begin{aligned} \sup_{|t| \leq \varepsilon} |Q'''_{s,r}(t)| &\leq C_\lambda (2s+r)^3 \sup_{|t| \leq \varepsilon} q(\xi+ts)q(\xi+t(s+r)) \\ &\leq C_\lambda (1+r)^3 e^{2M\varepsilon(1+r)} q(\xi)^2. \end{aligned}$$

Choose $\varepsilon_\lambda > 0$ so that $2M\varepsilon_\lambda \leq 1/2$. For $0 < \varepsilon \leq \varepsilon_\lambda$, the contribution of $R_{s,r}$ to $I_+ - I_-$ is bounded in absolute value by

$$\begin{aligned} \varepsilon^2 \int_0^\infty \int_0^{h(r)} |R_{s,r}| \, ds \, dr &\leq C_\lambda \varepsilon^5 q(\xi)^2 \int_0^\infty h(r)(1+r)^3 e^{2M\varepsilon(1+r)} \, dr \\ &\leq C_\lambda \varepsilon^5 q(\xi)^2 \int_0^\infty (1+r)^3 e^{-r/2} \, dr \\ &\leq C_\lambda \varepsilon^5 q(\xi). \end{aligned}$$

Here the second inequality uses $h(r) \leq e^{-r}$, with the constant $e^{1/2}$ absorbed into C_λ , and the last one follows from (4.20).

The linear term equals

$$2\varepsilon^3 q(\xi)q'(\xi) \int_0^\infty \int_0^{h(r)} (2s+r) \, ds \, dr = a\varepsilon^3 q(\xi)q'(\xi)$$

by Lemma 4.1. Together with the preceding remainder estimate, this proves (4.15).

The same relative estimate applied to the sum of the two products, now with the prefactor ε^2 , gives

$$I_+ + I_- \leq C_\lambda \varepsilon^2 q(\xi)^2 \int_0^\infty e^{-r/2} \, dr \leq C_\lambda \varepsilon^2 q(\xi),$$

and proves (4.16). $\square$

Proof of Lemma 4.5. Fix $x \geq \kappa\varepsilon^2 \geq 0$. Let Z_1, Z_2 be independent with CDF F_ε , and set $\tilde{Z}_i = Z_i \vee 0$. Then $\tilde{Z}_1, \tilde{Z}_2$ are independent with CDF G_ε . Since F_ε is continuous,

$$\mathbb{P}(Z_i < 0) = F_\varepsilon(0) = \alpha_\varepsilon.$$

For the parallel-composition map, the two input pairs agree when $Z_1, Z_2 \geq 0$. If at least one original input is negative, then both $g_-(Z_1, Z_2)$ and $g_-(\tilde{Z}_1, \tilde{Z}_2)$ are negative. Therefore the indicators of the events $\{g_-(Z_1, Z_2) \leq x\}$ and $\{g_-(\tilde{Z}_1, \tilde{Z}_2) \leq x\}$ coincide for $x \geq 0$, and hence

$$P_{F_\varepsilon}(x) = P_{G_\varepsilon}(x).$$

For the series-composition map, coordinatewise monotonicity gives $g_+(Z_1, Z_2) \leq g_+(\tilde{Z}_1, \tilde{Z}_2)$. Thus $S_{F_\varepsilon}(x) - S_{G_\varepsilon}(x)$ is the probability that

$$g_+(Z_1, Z_2) \leq x < g_+(\tilde{Z}_1, \tilde{Z}_2).$$

If $Z_1, Z_2 < 0$, then $g_+(Z_1, Z_2) < \log 2$, whereas $g_+(\tilde{Z}_1, \tilde{Z}_2) = \log 2$. This case therefore contributes only when $0 \leq x < \log 2$, and its contribution is at most α_ε^2 .

Suppose next that $Z_1 < 0 \leq Z_2$. Since $g_+(Z_1, Z_2) \leq x < g_+(0, Z_2)$, we have $Z_2 < g_+(Z_1, Z_2) \leq x$, and hence $Z_2 \leq x$. If $0 \leq x < \log 2$, this gives $0 \leq Z_2 \leq x$. If $x \geq \log 2$, the additional inequality

$$x < g_+(0, Z_2) = \log(1 + e^{Z_2})$$

is equivalent to $\log(e^x - 1) < Z_2$. Define

$$\begin{aligned} d(0) &= 0, \\ d(x) &= \begin{cases} x, & 0 < x < \log 2, \\ -\log(1 - e^{-x}), & x \geq \log 2, \end{cases} \\ 0 &\leq d(x) \leq \log 2. \end{aligned}$$

Indeed, for $x \geq \log 2$ we have $\log(e^x - 1) = x - d(x)$ and $1 - e^{-x} \geq 1/2$, which gives $d(x) \leq \log 2$. Independence and the same argument with the two inputs exchanged now yield

$$0 \leq S_{F_\varepsilon}(x) - S_{G_\varepsilon}(x) \leq \alpha_\varepsilon^2 \mathbf{1}_{\{0 \leq x < \log 2\}} + 2\alpha_\varepsilon (F_\varepsilon(x) - F_\varepsilon(x - d(x))).$$

Put $\xi = z_\varepsilon + \varepsilon x$. Since $q > 0$, (4.4) gives $|(\log q)'| \leq M_\lambda$. Consequently,

$$\begin{aligned} F_\varepsilon(x) - F_\varepsilon(x - d(x)) &= \varepsilon \int_0^{d(x)} q(\xi - \varepsilon s) \, ds \\ &\leq \varepsilon d(x) e^{M_\lambda \varepsilon d(x)} q(\xi) \leq C_\lambda \varepsilon q(\xi). \end{aligned}$$

Moreover, $\Phi(z_\varepsilon) = \varepsilon^4 \rightarrow 0$, so that $z_\varepsilon + \varepsilon \log 2 \rightarrow -\infty$. The left-tail ratio in (4.5) therefore gives, for all sufficiently small ε ,

$$q(z_\varepsilon + \varepsilon x) \geq c_\lambda \Phi(z_\varepsilon + \varepsilon x) \geq c_\lambda \Phi(z_\varepsilon) = c_\lambda \alpha_\varepsilon, \quad 0 \leq x \leq \log 2.$$

Thus, for $0 \leq x \leq \log 2$, the inequality $\alpha_\varepsilon^2 \leq c_\lambda^{-1} \alpha_\varepsilon q(\xi)$ holds. Using also $p_+ \leq 1$, $\varepsilon \leq 1$, and the equality of the parallel-composition CDFs, we obtain

$$\begin{aligned} 0 &\leq \mathcal{T}_{p_+} F_\varepsilon(x) - \mathcal{T}_{p_+} G_\varepsilon(x) = p_+ (S_{F_\varepsilon}(x) - S_{G_\varepsilon}(x)) \\ &\leq \alpha_\varepsilon^2 \mathbf{1}_{\{0 \leq x < \log 2\}} + C_\lambda \alpha_\varepsilon \varepsilon q(\xi) \leq C_\lambda \alpha_\varepsilon q(\xi) = C_\lambda \varepsilon^4 q(z_\varepsilon + \varepsilon x). \end{aligned}$$

This proves (4.18). □

4.3. Lower bound

Fix $0 < \lambda < \lambda_*$ and $0 < \kappa < \kappa_\lambda$. It is enough to construct an integrable Z_ε with CDF H_ε such that

$$F_0 \geq H_\varepsilon, \tag{4.21}$$

$$\mathcal{T}_{p_-} H_\varepsilon(x) \geq H_\varepsilon(x + \kappa \varepsilon^2), \quad x \in \mathbb{R}. \tag{4.22}$$

Indeed, if such H_ε exists, the same induction gives

$$F_n^-(x) \geq H_\varepsilon(x + n\kappa \varepsilon^2), \quad X_n(p_-) \leq_{\text{st}} Z_\varepsilon - n\kappa \varepsilon^2.$$

By integrability and Theorem 1.1, $v(p_-) \leq -\kappa \varepsilon^2$. By Lemma 2.5, $v(p_+) = -v(p_-)$, and hence

$$v(p_+) \geq \kappa \varepsilon^2.$$

Since $\delta = \varepsilon^3$, this gives

$$\liminf_{\delta \downarrow 0} \frac{v(\frac{1}{2} + \delta)}{\delta^{2/3}} \geq \kappa.$$

Letting first $\kappa \uparrow \kappa_\lambda$ and then $\lambda \uparrow \lambda_*$, and using (4.1), gives the lower bound

$$\liminf_{\delta \downarrow 0} \frac{v(\frac{1}{2} + \delta)}{\delta^{2/3}} \geq 2\zeta(3)^{1/3} \lambda_*. \tag{4.23}$$

It remains to construct an integrable Z_ε whose CDF H_ε has the two properties in (4.21) and (4.22).

Step 1: Construct H_ε and verify (4.21) Let $\Psi = \Psi_\lambda$ and $q = \Psi'$ be as in Proposition 4.3, using the zero extension of Ψ , and define

$$H_\varepsilon(x) = \Psi(\varepsilon x), \quad x \in \mathbb{R}.$$

This is an absolutely continuous CDF supported on $[0, \infty)$, with density

$$\rho_\varepsilon(x) = \varepsilon q(\varepsilon x) \mathbf{1}_{\{x > 0\}}.$$

The density vanishes on $(-\infty, 0)$ but has the positive right limit $\rho_\varepsilon(0+) = \varepsilon q(0) > 0$. Therefore, $F_0 \geq H_\varepsilon$, thus verifying (4.21). Moreover, the finite-mean conclusion of Proposition 4.3 shows that the associated random variable Z_ε is integrable. Thus only (4.22) remains to be verified.

Step 2: Prove (4.22) We start with a required estimate for (4.22).

Lemma 4.6. *There are $C_\lambda < \infty$ and $\varepsilon_\lambda > 0$ such that, for $0 < \varepsilon < \varepsilon_\lambda$ and $x \geq 0$,*

$$I_+(H_\varepsilon; x) - I_-(H_\varepsilon; x) \geq a\varepsilon^3 q(\varepsilon x) q'(\varepsilon x) - C_\lambda \varepsilon^5 q(\varepsilon x). \quad (4.24)$$

For $x \geq 0$, Lemma 4.6, (4.7), and the nonnegative last term in (2.12) give a one-step gain

$$\mathcal{T}_{p_-} H_\varepsilon(x) - H_\varepsilon(x) \geq \kappa_\lambda \varepsilon^3 q(\varepsilon x) - C_\lambda \varepsilon^5 q(\varepsilon x),$$

whereas Taylor's theorem gives

$$H_\varepsilon(x + \kappa \varepsilon^2) - H_\varepsilon(x) = \kappa \varepsilon^3 q(\varepsilon x) + O_{\lambda, \kappa}(\varepsilon^6 q(\varepsilon x)).$$

The strict inequality $\kappa < \kappa_\lambda$ therefore gives the desired barrier on $[0, \infty)$. The remaining case $x < 0$ is handled separately in Proposition 4.7.

Proposition 4.7. *Fix $0 < \lambda < \lambda_*$ and $0 < \kappa < \kappa_\lambda$. For all sufficiently small ε ,*

$$\mathcal{T}_{p_-} H_\varepsilon(x) \geq H_\varepsilon(x + \kappa \varepsilon^2), \quad x \in \mathbb{R}.$$

Proposition 4.7 proves the remaining case, so H_ε has the two properties used above.

We now prove Lemma 4.6 and Proposition 4.7.

Proof of Lemma 4.6. Fix $x \geq 0$ and put $\xi = \varepsilon x$. Let $\bar{q}$ be the strictly positive C^3 extension from (4.9), and define

$$\begin{aligned} \bar{I}_+(x) &= \varepsilon^2 \int_0^\infty \int_0^{h(r)} \bar{q}(\xi + \varepsilon s) \bar{q}(\xi + \varepsilon(s+r)) \, ds \, dr, \\ \bar{I}_-(x) &= \varepsilon^2 \int_0^\infty \int_0^{h(r)} \bar{q}(\xi - \varepsilon s) \bar{q}(\xi - \varepsilon(s+r)) \, ds \, dr. \end{aligned}$$

The density ρ_ε of H_ε satisfies

$$0 \leq \rho_\varepsilon(y) \leq \varepsilon \bar{q}(\varepsilon y), \quad y \in \mathbb{R},$$

with equality for $y > 0$. Since $x + s$ and $x + s + r$ are nonnegative, the integrands defining $I_+(H_\varepsilon; x)$ and $\bar{I}_+(x)$ therefore agree almost everywhere, while the integrand defining $I_-(H_\varepsilon; x)$ is bounded above by the one defining $\bar{I}_-(x)$. Consequently,

$$I_+(H_\varepsilon; x) = \bar{I}_+(x), \quad I_-(H_\varepsilon; x) \leq \bar{I}_-(x), \quad (4.25)$$

and hence

$$I_+(H_\varepsilon; x) - I_-(H_\varepsilon; x) \geq \bar{I}_+(x) - \bar{I}_-(x).$$

For fixed $r \geq 0$ and $0 \leq s \leq h(r)$, set

$$\overline{Q}_{s,r}(t) = \overline{q}(\xi + ts) \overline{q}(\xi + t(s+r)).$$

Then

$$\overline{I}_+(x) - \overline{I}_-(x) = \varepsilon^2 \int_0^\infty \int_0^{h(r)} (\overline{Q}_{s,r}(\varepsilon) - \overline{Q}_{s,r}(-\varepsilon)) \, ds \, dr. \quad (4.26)$$

Applying Taylor's theorem at 0 to $t = \pm\varepsilon$ and subtracting gives

$$\begin{aligned} \overline{Q}_{s,r}(\varepsilon) - \overline{Q}_{s,r}(-\varepsilon) &= 2\varepsilon \overline{Q}'_{s,r}(0) + \overline{R}_{s,r} = 2\varepsilon(2s+r) \overline{q}(\xi) \overline{q}'(\xi) + \overline{R}_{s,r}, \\ |\overline{R}_{s,r}| &\leq \frac{\varepsilon^3}{3} \sup_{|t| \leq \varepsilon} |\overline{Q}'''_{s,r}(t)|. \end{aligned} \quad (4.27)$$

We next estimate the remainder. Write $\overline{M} = \overline{M}_\lambda$, and allow C_λ to change from line to line. Since $\overline{q} > 0$, the case $j = 1$ in (4.9) gives $|(\log \overline{q})'| \leq \overline{M}$, and hence

$$e^{-\overline{M}|y|} \overline{q}(z) \leq \overline{q}(z+y) \leq e^{\overline{M}|y|} \overline{q}(z), \quad z, y \in \mathbb{R}. \quad (4.28)$$

Moreover, since $\overline{q} = q$ on $[0, \infty)$ and q is the density of Ψ , we have

$$\int_0^\infty \overline{q}(u) \, du = 1.$$

Thus, for every $\xi \geq 0$, (4.28) gives

$$1 \geq \int_\xi^{\xi+1} \overline{q}(u) \, du \geq e^{-\overline{M}} \overline{q}(\xi),$$

and therefore

$$\overline{q}(\xi)^2 \leq e^{\overline{M}} \overline{q}(\xi), \quad \xi \geq 0. \quad (4.29)$$

Choose $\varepsilon_\lambda > 0$ so that $2\overline{M}\varepsilon_\lambda \leq 1/2$, and assume henceforth that $0 < \varepsilon < \varepsilon_\lambda$. Since $0 \leq s \leq h(r) \leq \log 2 < 1$, we have $2s+r \leq 2(1+r)$. Hence Leibniz's rule, (4.9), and (4.28) imply, for all $|t| \leq \varepsilon$,

$$\begin{aligned} |\overline{Q}'''_{s,r}(t)| &\leq C_\lambda \sum_{j=0}^3 \binom{3}{j} s^j (s+r)^{3-j} \overline{q}(\xi+ts) \overline{q}(\xi+t(s+r)) \\ &= C_\lambda (2s+r)^3 \overline{q}(\xi+ts) \overline{q}(\xi+t(s+r)) \\ &\leq C_\lambda (1+r)^3 e^{2\overline{M}\varepsilon(1+r)} \overline{q}(\xi)^2. \end{aligned}$$

The contribution of $\overline{R}_{s,r}$ to (4.26) satisfies

$$\begin{aligned} \left| \varepsilon^2 \int_0^\infty \int_0^{h(r)} \overline{R}_{s,r} \, ds \, dr \right| &\leq C_\lambda \varepsilon^5 \overline{q}(\xi)^2 \int_0^\infty h(r) (1+r)^3 e^{2\overline{M}\varepsilon(1+r)} \, dr \\ &\leq C_\lambda \varepsilon^5 \overline{q}(\xi)^2 \int_0^\infty (1+r)^3 e^{-r/2} \, dr \\ &\leq C_\lambda \varepsilon^5 \overline{q}(\xi). \end{aligned}$$

Here the second inequality uses $h(r) \leq e^{-r}$, with the factor $e^{1/2}$ absorbed into C_λ , and the last one uses (4.29).

The linear term in (4.27) contributes

$$2\varepsilon^3 \bar{q}(\xi) \bar{q}'(\xi) \int_0^\infty \int_0^{h(r)} (2s+r) \, ds \, dr = a\varepsilon^3 \bar{q}(\xi) \bar{q}'(\xi)$$

by Lemma 4.1. Consequently,

$$\left| \bar{I}_+(x) - \bar{I}_-(x) - a\varepsilon^3 \bar{q}(\xi) \bar{q}'(\xi) \right| \leq C_\lambda \varepsilon^5 \bar{q}(\xi). \quad (4.30)$$

Finally, since $\xi \geq 0$ and $\bar{q}$ agrees with q on $[0, \infty)$,

$$\bar{q}(\xi) = q(\xi), \quad \bar{q}'(\xi) = q'(\xi),$$

where $q'(0)$ denotes the right derivative $q'(0+)$. Combining (4.25) and (4.30), and recalling that $\xi = \varepsilon x$, yields

$$I_+(H_\varepsilon; x) - I_-(H_\varepsilon; x) \geq a\varepsilon^3 q(\varepsilon x) q'(\varepsilon x) - C_\lambda \varepsilon^5 q(\varepsilon x),$$

which proves (4.24). $\square$

Proof of Proposition 4.7. Put $s_\varepsilon = \kappa \varepsilon^2$. If $x < -s_\varepsilon$, the right-hand side is zero. The desired inequality is therefore immediate because $\mathcal{T}_{p_-} H_\varepsilon$ is a CDF.

Suppose $x \geq 0$ and write $\xi = \varepsilon x$. Since the last term in (2.12) is nonnegative when $p = p_-$, Lemma 4.6 and (4.7) give

$$\begin{aligned} \mathcal{T}_{p_-} H_\varepsilon(x) - H_\varepsilon(x) &\geq \varepsilon^3 (a q(\xi) q'(\xi) + 2\Psi(\xi)(1 - \Psi(\xi))) - C_\lambda \varepsilon^5 q(\xi) \\ &= \kappa_\lambda \varepsilon^3 q(\xi) - C_\lambda \varepsilon^5 q(\xi). \end{aligned}$$

On the other hand, by (4.8) and Taylor's theorem,

$$H_\varepsilon(x + s_\varepsilon) - H_\varepsilon(x) = \kappa \varepsilon^3 q(\xi) + O_{\lambda, \kappa}(\varepsilon^6 q(\xi)).$$

Because $\kappa < \kappa_\lambda$, the desired inequality follows for all $x \geq 0$ when ε is small.

It remains to consider $-s_\varepsilon \leq x < 0$. Here $H_\varepsilon(x) = 0$ and $I_-(H_\varepsilon; x) = 0$, so

$$\mathcal{T}_{p_-} H_\varepsilon(x) = (1 + 2\varepsilon^3) I_+(H_\varepsilon; x) \geq I_+(H_\varepsilon; x).$$

Let $q_0 = q(0+) > 0$. Since $q(z) \rightarrow q_0$ as $z \downarrow 0$, choose $\eta > 0$ such that $q(z) \geq \frac{q_0}{2}$ for $0 \leq z \leq \eta$. Set $b = h(1)/4$, and take ε sufficiently small that

$$s_\varepsilon < b/2, \quad \varepsilon(3b+1) \leq \eta.$$

Since h is decreasing, for $0 \leq r \leq 1$ we have $h(r) \geq h(1) = 4b$, so the rectangle

$$0 \leq r \leq 1, \quad 2b \leq s \leq 3b$$

is contained in the integration region $\{r \geq 0, 0 \leq s \leq h(r)\}$. Moreover, for $-s_\varepsilon \leq x < 0$ and (r, s) in this rectangle,

$$0 < 2b - s_\varepsilon \leq x + s \leq x + s + r \leq 3b + 1.$$

It follows that

$$\rho_\varepsilon(x + s) \geq \frac{\varepsilon q_0}{2}, \quad \rho_\varepsilon(x + s + r) \geq \frac{\varepsilon q_0}{2}.$$

Restricting $I_+(H_\varepsilon; x)$ to this rectangle therefore gives

$$\begin{aligned} I_+(H_\varepsilon; x) &\geq \int_0^1 \int_{2b}^{3b} \rho_\varepsilon(x + s) \rho_\varepsilon(x + s + r) \, ds \, dr \\ &\geq \int_0^1 \int_{2b}^{3b} \frac{\varepsilon^2 q_0^2}{4} \, ds \, dr = \frac{b q_0^2}{4} \varepsilon^2. \end{aligned} \quad (4.31)$$

Since $q(z) \rightarrow q_0$ as $z \downarrow 0$, we have $q(z) \leq 2q_0$ for all sufficiently small $z > 0$. Moreover, on the present strip, $0 \leq x + s_\varepsilon \leq s_\varepsilon$. Hence, for all sufficiently small ε ,

$$H_\varepsilon(x + s_\varepsilon) = \Psi(\varepsilon(x + s_\varepsilon)) = \int_0^{\varepsilon(x+s_\varepsilon)} q(z) \, dz \leq 2q_0 \varepsilon(x + s_\varepsilon) \leq 2q_0 \varepsilon s_\varepsilon = 2\kappa q_0 \varepsilon^3.$$

The lower bound (4.31) dominates this quantity, completing the proof. $\square$

Proof of Theorem 1.3. The existence, positivity, and minimality of λ_* for the boundary-value problem (1.5) follow from Proposition A.3. Combining (4.14) and (4.23), and using Lemma 2.5, yields (1.6). For the first-moment assertions, Theorem 1.2 gives $\gamma_R(1/2 + \delta) = v_R(1/2 + \delta)$. On the subcritical side, since $\delta^{2/3} \gg \delta$, (1.6), Theorem 1.2, and $\log(1 - 2\delta) \sim -2\delta$ give, for all sufficiently small $\delta > 0$,

$$\gamma_R\left(\frac{1}{2} - \delta\right) = v_R\left(\frac{1}{2} - \delta\right) \vee \log(1 - 2\delta) = \log(1 - 2\delta).$$

Together with (1.6), these identities yield (1.7). This completes the proof. $\square$

Remark 4.8 (Why the exponent $2/3$?). The exponent $2/3$ can be inferred from the orders of the terms in the one-step recursion. To study the scaling of $v_R(\frac{1}{2} + \delta) = \lim_{n \rightarrow \infty} n^{-1} X_n(\frac{1}{2} + \delta)$ as $\delta \rightarrow 0$, we compare $\mathcal{T}_{1/2+\delta} F_n^+(\cdot) - F_n^+(\cdot) = F_{n+1}^+(\cdot) - F_n^+(\cdot)$ with the change $F_n^+(\cdot - s) - F_n^+(\cdot)$ induced by a translation of size s , to be determined. If one can choose s , independently of n , such that $F_{n+1}^+(\cdot) - F_n^+(\cdot) \asymp F_n^+(\cdot - s) - F_n^+(\cdot)$, then heuristically letting $n \rightarrow \infty$ gives $v_R(\frac{1}{2} + \delta) \asymp s$.

To implement this idea, we seek an auxiliary CDF K satisfying $\mathcal{T}_{1/2+\delta} K(\cdot) - K(\cdot) \asymp K(\cdot - s) - K(\cdot)$ and approximating F_0^+ . This suggests $F_n^+(\cdot) \asymp K(\cdot - ns)$ and hence $v_R(\frac{1}{2} + \delta) \asymp s$. The construction is guided by the following heuristic: Let $K_\eta(x) = K(\eta x)$ be a smooth CDF of width η^{-1} . At criticality, the first non-vanishing term in $\mathcal{T}_{1/2} K_\eta - K_\eta$ is of order η^3 , whereas the bias $\mathcal{T}_{1/2+\delta} K_\eta(\cdot) - \mathcal{T}_{1/2} K_\eta(\cdot)$ produces a term of order δ . Balancing these contributions gives $\eta^3 \asymp \delta$. The change $K_\eta(\cdot - s) - K_\eta(\cdot)$ caused by translation through s has order $s\eta$; matching this with the one-step change of order δ gives $s\eta \asymp \delta$. Thus $\eta \asymp \delta^{1/3}$ and $s \asymp \delta^{2/3}$, which predicts the scale of the resistance speed.

A. ODE analysis and admissible parameters

In this appendix, we study the ordinary differential equation (ODE) for $W \in C([0, 1]) \cap C^1((0, 1))$ with parameter $\lambda > 0$,

$$W^2 W' - \lambda W + u(1 - u) = 0, \quad 0 < u < 1, \quad (\text{A.1})$$

subject to

$$W(0) = W(1) = 0, \quad W > 0 \quad \text{on } (0, 1). \quad (\text{A.2})$$

We prove that there exists a least admissible parameter $\lambda_* > 0$: the boundary-value problem has a unique positive solution for every $\lambda \geq \lambda_*$ and no such solution for $0 < \lambda < \lambda_*$. We then determine the possible endpoint asymptotics of these solutions at $u = 0$.

A.1. Admissible parameters of the ODE

For $t \in [0, 1]$, let $y(t) = (W(1 - t))^3$. Under this substitution, (A.1) is equivalent to

$$y' = 3t(1 - t) - 3\lambda y^{1/3}, \quad 0 < t < 1. \quad (\text{A.3})$$

The boundary conditions (A.2) are equivalent to

$$y(0) = y(1) = 0, \quad y > 0 \quad \text{on } (0, 1). \quad (\text{A.4})$$

We first study (A.3) with the initial condition $y(0) = 0$:

$$y' = 3t(1 - t) - 3\lambda y^{1/3}, \quad 0 < t < 1, \quad y(0) = 0. \quad (\text{A.5})$$

We prove existence and uniqueness for (A.5), and then determine the parameters $\lambda > 0$ for which (A.4) also holds.

Local existence for (A.5) follows from Peano's theorem, since the right-hand side is continuous. To obtain existence on all of $[0, 1]$ and uniqueness, we use the following comparison principle; this is needed because the cube-root term is not locally Lipschitz at zero.

Lemma A.1. *Let $I = [t_0, t_1]$, $a \in C(I)$ and $b > 0$. Suppose that $\underline{y}, \bar{y} \in C(I) \cap C^1(\overset{\circ}{I})$ satisfy*

$$\underline{y}' \leq a(t) - b\underline{y}^{1/3}, \quad \bar{y}' \geq a(t) - b\bar{y}^{1/3}.$$

If $\underline{y}(t_0) \leq \bar{y}(t_0)$, then $\underline{y} \leq \bar{y}$ on I .

Proof. Let $d := \underline{y} - \bar{y}$ and $\Psi(s) = \frac{1}{2}(s_+)^2$. For every $t \in \overset{\circ}{I}$, the chain rule gives

$$\frac{d}{dt}\Psi(d(t)) \leq -bd_+(t)(\underline{y}(t)^{1/3} - \bar{y}(t)^{1/3}) \leq 0.$$

Thus $\Psi(d)$ is non-increasing on every compact subinterval of $\overset{\circ}{I}$. By continuity at the endpoints and $\Psi(d(t_0)) = 0$, it follows that $\Psi(d) = 0$ throughout I , and hence $\underline{y} \leq \bar{y}$ on I . $\square$

Applying the comparison principle, we have the following lemma.

Lemma A.2. *For every $\lambda > 0$, equation (A.5) has a unique solution y_λ on $[0, 1]$, and*

$$y_\lambda(t) > 0, \quad 0 < t < 1. \quad (\text{A.6})$$

If $\lambda_2 > \lambda_1$, then

$$y_{\lambda_2}(t) < y_{\lambda_1}(t), \quad 0 < t < 1. \quad (\text{A.7})$$

Finally, $\lambda \mapsto y_\lambda$ is continuous from $(0, \infty)$ into $C([0, 1])$. More precisely, for all $\lambda_1, \lambda_2 > 0$,

$$\|y_{\lambda_1} - y_{\lambda_2}\|_{C([0, 1])} \leq 3 \left(\frac{3}{4}\right)^{1/3} |\lambda_1 - \lambda_2|.$$

Proof. We first prove existence and uniqueness of the solution on $[0, 1]$. By Peano's theorem, the ODE has at least a local solution. Since the constant zero is a subsolution, Lemma A.1 gives $y_\lambda \geq 0$ on its interval of existence. Moreover,

$$y'_\lambda(t) \leq 3t(1 - t) \leq \frac{3}{4},$$

so the solution remains bounded in $[0, 1]$ and therefore extends to $[0, 1]$. Applying the comparison lemma in both directions proves uniqueness.

We then prove (A.6). If $y_\lambda(t_0) = 0$ for some $t_0 \in (0, 1)$, then t_0 is an interior minimum because $y_\lambda \geq 0$, so $y'_\lambda(t_0) = 0$. But the equation gives $y'_\lambda(t_0) = 3t_0(1 - t_0) > 0$, a contradiction.

We next show (A.7). For $\lambda_2 > \lambda_1$, the function y_{λ_1} is a supersolution for the equation with parameter λ_2 , so $y_{\lambda_2} \leq y_{\lambda_1}$. If $y_{\lambda_2}(t_0) = y_{\lambda_1}(t_0)$ for some $t_0 \in (0, 1)$, then t_0 is an interior minimum of $y_{\lambda_1} - y_{\lambda_2}$, so $y'_{\lambda_1}(t_0) - y'_{\lambda_2}(t_0) = 0$. Substituting the equality at t_0 into the equation gives

$$y'_{\lambda_1}(t_0) - y'_{\lambda_2}(t_0) = 3(\lambda_2 - \lambda_1)y_{\lambda_1}^{1/3}(t_0) > 0,$$

a contradiction.

Finally, we prove continuity of $\lambda \mapsto y_\lambda$. For every $\lambda > 0$, since $y_\lambda(0) = 0$ and $y'_\lambda \leq 3/4$, we have the uniform bound

$$0 \leq y_\lambda(t) \leq \frac{3}{4}, \quad 0 \leq t \leq 1.$$

Take $0 < \lambda_1 \leq \lambda_2$. The parameter monotonicity already proved implies $0 \leq y_{\lambda_2} \leq y_{\lambda_1}$. Hence, for $0 < t < 1$,

$$\begin{aligned} (y_{\lambda_1} - y_{\lambda_2})'(t) &= -3\lambda_1(y_{\lambda_1}(t)^{1/3} - y_{\lambda_2}(t)^{1/3}) + 3(\lambda_2 - \lambda_1)y_{\lambda_2}(t)^{1/3} \\ &\leq 3\left(\frac{3}{4}\right)^{1/3}(\lambda_2 - \lambda_1). \end{aligned}$$

Since $y_{\lambda_1}(0) = y_{\lambda_2}(0) = 0$,

$$0 \leq y_{\lambda_1}(t) - y_{\lambda_2}(t) \leq 3\left(\frac{3}{4}\right)^{1/3}(\lambda_2 - \lambda_1).$$

Interchanging λ_1 and λ_2 when necessary and taking the supremum over $t \in [0, 1]$ proves the asserted Lipschitz estimate. In particular, $\lambda \mapsto y_\lambda$ is continuous as a map into $C([0, 1])$. $\square$

For later comparison constructions, it is convenient to introduce the following function. For a nonnegative function w with $w(0) = 0$, set

$$\mathcal{R}_\lambda[w](t) = w(t)^2 w'(t) + \lambda w(t) - t(1 - t).$$

If $z = w^3$, then

$$z' - (3t(1 - t) - 3\lambda z^{1/3}) = 3\mathcal{R}_\lambda[w].$$

Thus $\mathcal{R}_\lambda[w] \leq 0$ corresponds to a subsolution of (A.5), and $\mathcal{R}_\lambda[w] \geq 0$ to a supersolution.

Define the admissible set

$$\mathcal{A} := \{\lambda > 0 : y_\lambda(1) = 0\}.$$

By Lemma A.2, $\lambda \in \mathcal{A}$ if and only if

$$W_\lambda(u) = [y_\lambda(1 - u)]^{1/3}$$

solves (A.1) and (A.2); for a fixed λ , this positive solution is unique.

Proposition A.3. *There exists $\lambda_* \in (0, \infty)$ such that*

$$\mathcal{A} = [\lambda_*, \infty). \quad (\text{A.8})$$

Moreover, λ_* satisfies the bounds

$$\left(\frac{4}{15\sqrt{2}}\right)^{2/3} \leq \lambda_* \leq \frac{\sqrt{3}}{2}. \quad (\text{A.9})$$

Proof. We first show that $\mathcal{A}$ is nonempty. Take $\bar{w}(t) := \frac{3}{2\lambda}t(1 - t)$. Then

$$\mathcal{R}_\lambda[\bar{w}] = t(1 - t) \left(\frac{1}{2} + \left(\frac{3}{2\lambda} \right)^3 t(1 - t)(1 - 2t) \right).$$

Notice

$$\max_{0 \leq t \leq 1} \{-t(1 - t)(1 - 2t)\} = \frac{1}{6\sqrt{3}},$$

so if $\lambda \geq \sqrt{3}/2$, we have

$$\frac{1}{2} + \left(\frac{3}{2\lambda} \right)^3 t(1 - t)(1 - 2t) \geq 0.$$

Hence $\mathcal{R}_\lambda[\bar{w}] \geq 0$ on $[0, 1]$. Therefore $\bar{y} := \bar{w}^3$ is a supersolution of (A.5). Since $\bar{y}(1) = 0$, the comparison principle (Lemma A.1) yields $y_\lambda(1) = 0$. This is equivalent to $\lambda \in \mathcal{A}$.

By Lemma A.2, y_λ is monotone in λ . If $\lambda_1 \in \mathcal{A}$ and $\lambda_2 > \lambda_1$, then

$$0 \leq y_{\lambda_2}(1) \leq y_{\lambda_1}(1) = 0,$$

so $\lambda_2 \in \mathcal{A}$. Thus $\mathcal{A}$ is upward closed. Let $\lambda_* := \inf \mathcal{A}$.

We next prove that $\lambda_* > 0$. Fix $\lambda \in \mathcal{A}$. Dividing (A.1) by W_λ for $u \in (0, 1)$ gives

$$(W_\lambda^2(u))' = 2\lambda - 2 \frac{u(1-u)}{W_\lambda(u)} \leq 2\lambda. \quad (\text{A.10})$$

Since $W_\lambda(0) = 0$, direct integration gives $W_\lambda(u)^2 \leq 2\lambda u$, that is, $W_\lambda(u) \leq \sqrt{2\lambda u}$ for $0 \leq u \leq 1$.

To justify the endpoint integral, integrate the identity in (A.10) over $[\rho, 1 - \rho]$, where $0 < \rho < 1/2$:

$$W_\lambda(1 - \rho)^2 - W_\lambda(\rho)^2 = 2\lambda(1 - 2\rho) - 2 \int_\rho^{1-\rho} \frac{u(1-u)}{W_\lambda(u)} du.$$

As $\rho \downarrow 0$, the left-hand side tends to zero by continuity and the boundary conditions. The integral converges by monotone convergence, and therefore

$$\lambda = \int_0^1 \frac{u(1-u)}{W_\lambda(u)} du. \quad (\text{A.11})$$

Substituting $W_\lambda(u) \leq \sqrt{2\lambda u}$ into (A.11) gives

$$\lambda \geq \frac{1}{\sqrt{2\lambda}} \int_0^1 u^{1/2}(1-u) du = \frac{4}{15\sqrt{2\lambda}}.$$

This proves the lower bound in (A.9), and hence $\lambda_* > 0$.

Finally, Lemma A.2 shows that $\lambda \mapsto y_\lambda$ is continuous on $(0, \infty)$. Since $\lambda_* > 0$, a sequence in $\mathcal{A}$ decreasing to λ_* gives $y_{\lambda_*}(1) = 0$, so $\lambda_* \in \mathcal{A}$. Combining this with the upward-closed property proves $\mathcal{A} = [\lambda_*, \infty)$ and completes (A.8). $\square$

A.2. Asymptotics of W_λ at endpoints

For $0 < \lambda < \lambda_*$, Lemma A.2 and Proposition A.3 imply that $y_\lambda(1) > 0$. Hence the solution W_λ satisfying $W_\lambda(1) = 0$ has $W_\lambda(0) > 0$ and does not satisfy both boundary conditions in (A.2). For $\lambda \geq \lambda_*$, the boundary-value problem has a unique positive solution. We now determine the endpoint asymptotics in these two regimes.

Proposition A.4. *If $0 < \lambda < \lambda_*$, then*

$$y_\lambda(1) > 0, \quad W_\lambda(u) := [y_\lambda(1-u)]^{1/3} > 0 \quad \text{for } 0 \leq u < 1, \quad (\text{A.12})$$

with $W_\lambda(1) = 0$ and $W_\lambda(0) > 0$. The function W_λ solves (A.1) on $(0, 1)$ and is smooth at $u = 0$.

Proof. Equation (A.12) is a direct corollary of Lemma A.2 and Proposition A.3.

It remains to prove that W_λ is smooth at $u = 0$. For $u \in [0, 1]$, we have $W_\lambda(u) > 0$. Dividing both sides of (A.1) by $W_\lambda(u)^2$ gives

$$W_\lambda'(u) = \frac{\lambda W_\lambda(u) - u(1-u)}{W_\lambda(u)^2} =: F(u, W_\lambda(u)).$$

Since W_λ is continuous and $W_\lambda(0) > 0$, there are $c > 0$ and $\eta > 0$ such that $W_\lambda(u) \geq c$ for $0 \leq u \leq \eta$. The function

$$F(u, w) = \frac{\lambda w - u(1-u)}{w^2}$$

is smooth on a neighborhood of $\{(u, W_\lambda(u)) : u \in [0, \eta]\}$. In particular,

$$W'_\lambda(u) = F(u, W_\lambda(u)) \longrightarrow F(0, W_\lambda(0)) = \frac{\lambda}{W_\lambda(0)} \quad \text{as } u \downarrow 0.$$

The derivative is therefore bounded near zero. Integrating first over $[\delta, u]$ and then letting $\delta \downarrow 0$ yields

$$W_\lambda(u) - W_\lambda(0) = \int_0^u F(s, W_\lambda(s)) \, ds.$$

Thus W_λ extends to a C^1 function at zero, with $W'_\lambda(0) = \lambda/W_\lambda(0)$. Finally, if W_λ is C^k on $[0, \eta]$, then the smoothness of F shows that $W'_\lambda = F(\cdot, W_\lambda)$ is C^k , and hence W_λ is C^{k+1} . Induction proves that W_λ is smooth at $u = 0$. $\square$

Lemma A.5. *For every $\lambda > 0$,*

$$W_\lambda(u) \sim \frac{1-u}{\lambda}, \quad u \uparrow 1.$$

Proof. Set

$$w_\lambda(t) := W_\lambda(1-t) = y_\lambda(t)^{1/3}, \quad 0 \leq t < 1.$$

Thus it suffices to prove that $w_\lambda(t)/t \rightarrow 1/\lambda$ as $t \downarrow 0$. Fix constants $0 < k_- < \frac{1}{\lambda} < k_+$, and define $\underline{w}(t) = k_-t$, $\overline{w}(t) = k_+t$. By direct computation, for any $k > 0$,

$$\mathcal{R}_\lambda[kt] = t(\lambda k - 1) + t^2(k^3 + 1).$$

Since $\lambda k_- - 1 < 0$ and $\lambda k_+ - 1 > 0$, there exists a sufficiently small $\delta > 0$ such that $\underline{y} := \underline{w}^3$ and $\overline{y} := \overline{w}^3$ are a subsolution and a supersolution, respectively, of (A.5) on $[0, \delta]$. All three functions have the same initial value 0, so Lemma A.1 gives

$$(k_-t)^3 \leq y_\lambda(t) \leq (k_+t)^3, \quad 0 \leq t \leq \delta.$$

Then it follows that

$$k_- \leq \liminf_{t \downarrow 0} \frac{w_\lambda(t)}{t} \leq \limsup_{t \downarrow 0} \frac{w_\lambda(t)}{t} \leq k_+.$$

Letting $k_- \uparrow 1/\lambda$ and $k_+ \downarrow 1/\lambda$ yields $w_\lambda(t)/t \rightarrow 1/\lambda$ as $t \downarrow 0$. Setting $u = 1 - t$ gives the desired result. $\square$

We next classify the asymptotics at $u = 0$ for $\lambda \in \mathcal{A}$.

Lemma A.6. *If $\lambda \in \mathcal{A}$, exactly one of the following asymptotics holds:*

$$W_\lambda(u) \sim \frac{u}{\lambda}, \quad u \downarrow 0, \tag{A.13}$$

$$W_\lambda(u) \sim \sqrt{2\lambda u}, \quad u \downarrow 0. \tag{A.14}$$

Proof. Write $W = W_\lambda$ and

$$H(u) = \lambda W(u) - u(1-u) = W(u)^2 W'(u).$$

Since $W(0) = 0 < W(u)$, the mean-value theorem shows that $H > 0$ at least on a sequence of points tending to zero. At any zero $v \in (0, 1/2)$ of H , $W'(v) = 0$ and

$$H'(v) = -(1-2v) < 0.$$

Thus H has at most one zero in $(0, 1/2)$. Indeed, after a zero, H is negative locally, whereas at the first return to zero its derivative would have to be nonnegative. Hence $H > 0$ in a sufficiently small interval $(0, u_0]$, and W is strictly increasing on $[0, u_0]$.

Consequently, W has a continuous inverse $U : [0, x_0] \rightarrow [0, u_0]$, where $x_0 = W(u_0)$. Moreover,

$$U \in C([0, x_0]) \cap C^1((0, x_0]), \quad U(0) = 0, \quad U(x) > 0 \quad (0 < x \leq x_0).$$

Let $r(x) := U(x)/x$. The original equation (A.1) gives, for $x \in (0, x_0)$,

$$\begin{aligned} U'(x) &= \frac{x^2}{\lambda x - U(x) + U(x)^2} = \frac{x}{\lambda - r(x) + xr(x)^2}, \\ xr'(x) &= \frac{x}{\lambda - r(x) + xr(x)^2} - r(x). \end{aligned} \tag{A.15}$$

Since $U'(x) > 0$, the denominator $\lambda - r(x) + xr(x)^2$ is positive. If $r(x_j) \geq 2\lambda$ for some $x_j \downarrow 0$, then

$$x_j r(x_j)^2 > r(x_j) - \lambda \geq \frac{1}{2} r(x_j), \quad U(x_j) = x_j r(x_j) > \frac{1}{2},$$

contrary to $U(x) \rightarrow 0$. Thus r is bounded near zero. Since $r < \lambda + xr^2$, we also have

$$\limsup_{x \downarrow 0} r(x) \leq \lambda. \tag{A.16}$$

Let $\rho(s) := r(e^{-s})$, $s \in [-\log x_0, \infty)$. Equation (A.15) becomes

$$\rho'(s) = \rho(s) - \frac{e^{-s}}{\lambda - \rho(s) + e^{-s}\rho(s)^2}. \tag{A.17}$$

We show that either $\lim_{s \rightarrow \infty} \rho(s) = 0$ or $\lim_{s \rightarrow \infty} \rho(s) = \lambda$.

We use the following elementary observation. Suppose that there exists a sufficiently large S such that $\phi'(s) > 0$ whenever $s \geq S$ and $\phi(s) = a$. Then, for every $s_0 \geq S$ with $\phi(s_0) \geq a$, one has

$$\phi(s) \geq a \quad \text{for all } s \geq s_0.$$

Indeed, otherwise at the first time when ϕ falls below a , its derivative at the boundary level a would be non-positive.

Suppose first that $\liminf \rho = 0$. If $\limsup \rho > 0$, choose $0 < \eta < \min\{\lambda/2, \limsup \rho\}$. At $\rho(s) = \eta$, for all large s ,

$$\rho'(s) \geq \eta - \frac{2e^{-s}}{\lambda} \geq \frac{\eta}{2} > 0.$$

Since $\limsup \rho > \eta$, the observation gives $\rho \geq \eta$ eventually, contrary to $\liminf \rho = 0$. Hence $\rho(s) \rightarrow 0$.

Now suppose $L := \liminf \rho > 0$. Fix $\delta \in (0, \lambda)$ and choose $\eta \in (0, L/2)$. There exists S such that, for every $s \geq S$,

$$\rho(s) \geq \eta, \quad \frac{e^{-s}}{\delta} \leq \frac{\eta}{2}.$$

Hence, whenever $s \geq S$ and $\rho(s) \leq \lambda - \delta$, (A.17) gives

$$\rho'(s) \geq \eta - \frac{e^{-s}}{\delta} \geq \frac{\eta}{2}.$$

This lower bound is uniform for $s \geq S$ and $\rho(s) \leq \lambda - \delta$. Hence ρ reaches $\lambda - \delta$ in finite time. The preceding observation then gives $\rho(s) \geq \lambda - \delta$ for all sufficiently large s . Therefore

$$\liminf_{s \rightarrow \infty} \rho(s) \geq \lambda - \delta.$$

Since $\delta > 0$ is arbitrary, together with (A.16) we have $\rho(s) \rightarrow \lambda$.

Therefore, we have either $r(x) \rightarrow \lambda$ or $r(x) \rightarrow 0$, as $x \rightarrow 0$. The first case says $u/W(u) \rightarrow \lambda$ and gives (A.13). In the second case, we have $U'(x) \sim x/\lambda$, which implies $U(x) \sim x^2/(2\lambda)$, and this gives (A.14). $\square$

Using Lemma A.6, we now determine the asymptotics of $W_\lambda(u)$ at $u = 0$ for every $\lambda \in \mathcal{A}$.

Proposition A.7. *At $\lambda = \lambda_*$, we have*

$$W_{\lambda_*}(u) \sim \sqrt{2\lambda_*}u, \quad u \downarrow 0. \quad (\text{A.18})$$

For every $\lambda > \lambda_$, we have*

$$W_{\lambda}(u) \sim \frac{u}{\lambda}, \quad u \downarrow 0. \quad (\text{A.19})$$

Proof. We first show that an admissible parameter $\lambda \in \mathcal{A}$ with linear asymptotics cannot be the left endpoint $\lambda_* \in \mathcal{A}$. Suppose $\lambda_0 \in \mathcal{A}$ and $W_{\lambda_0}(u) \sim u/\lambda_0$ as $u \downarrow 0$. Choose some $k > 1/\lambda_0$. Then $\lambda_0 k - 1 > 0$. Choose a small $\eta > 0$ such that

$$W_{\lambda_0}(\eta) < k\eta, \quad \eta|1 - k^3| \leq \frac{\lambda_0 k - 1}{4}.$$

Fix this k and η . By Lemma A.2, the map $\lambda \mapsto W_{\lambda}(\eta)$ is continuous, so we can choose $\lambda' < \lambda_0$ sufficiently close to λ_0 such that

$$\lambda'k - 1 \geq \frac{\lambda_0 k - 1}{2}, \quad W_{\lambda'}(\eta) < k\eta.$$

Now consider $w(t) = k(1 - t)$ on $t \in [1 - \eta, 1]$. Writing $u = 1 - t$, we have

$$\mathcal{R}_{\lambda'}[w] = u(\lambda'k - 1) + u^2(1 - k^3) \geq \frac{\lambda_0 k - 1}{4}u > 0.$$

At $t = 1 - \eta$, since $W_{\lambda'}(\eta) < k\eta$, the comparison principle gives

$$0 \leq y_{\lambda'}(t) \leq [k(1 - t)]^3, \quad 1 - \eta \leq t \leq 1.$$

In particular, $y_{\lambda'}(1) = 0$, so $\lambda' \in \mathcal{A}$. We have therefore found an admissible parameter strictly smaller than λ_0 . Thus λ_* cannot have linear asymptotics. Otherwise, one could find $\lambda' < \lambda_*$ with $\lambda' \in \mathcal{A}$, contradicting the definition of λ_* . The dichotomy in Lemma A.6 then gives (A.18).

We then prove (A.19). If $\lambda > \lambda_*$, Lemma A.2 gives $W_{\lambda}(u) < W_{\lambda_*}(u)$ for $0 < u < 1$. If $W_{\lambda}(u) \sim \sqrt{2\lambda}u$, then

$$\frac{W_{\lambda}(u)}{W_{\lambda_*}(u)} \longrightarrow \sqrt{\frac{\lambda}{\lambda_*}} > 1$$

as $u \downarrow 0$, a contradiction. Thus (A.19) holds. $\square$

B. Proofs of Propositions 4.2 and 4.3

This appendix has three components: Lemma B.1 provides an abstract finite asymptotic expansion, Lemma B.2 transfers it to the linear endpoint branches of W_{λ} , and the final two proofs convert those endpoint estimates into the two CDFs required in Propositions 4.2 and 4.3.

Lemma B.1. *Let $I \subset \mathbb{R}$ be an open interval containing y_* , and let $F, G \in C^9(I)$ satisfy*

$$F(y_*) = 0, \quad \mu := F'(y_*) \neq 0.$$

Suppose that, for some $T_0 > 0$, a function $Y \in C^1([T_0, \infty); I)$ satisfies

$$Y'(t) = F(Y(t)) + \frac{1}{t}G(Y(t)), \quad Y(t) \longrightarrow y_* \quad \text{as } t \rightarrow \infty. \quad (\text{B.1})$$

Then there exist coefficients $b_1, \dots, b_4$, with

$$b_1 = -\frac{G(y_*)}{F'(y_*)},$$

such that, for $0 \leq \ell \leq 4$,

$$\frac{d^{\ell}}{dt^{\ell}} \left(Y(t) - y_* - \sum_{j=1}^4 b_j t^{-j} \right) = O(t^{-5-\ell}) \quad \text{as } t \rightarrow \infty. \quad (\text{B.2})$$

Proof. Equation (B.1) and the regularity of F and G imply, by induction, that Y has all derivatives required below on (T_0, ∞) . We first construct an expansion with eight terms. Let

$$\begin{aligned} p_8(x) &= \sum_{j=1}^8 b_j x^j, \\ P_8(t) &= y_* + p_8(t^{-1}), \\ D_8(t) &= P'_8(t) - F(P_8(t)) - t^{-1}G(P_8(t)). \end{aligned}$$

Here D_8 is the defect of P_8 . With $x = t^{-1}$, this defect can be written as $D_8(t) = H_8(x)$, where

$$H_8(x) = -x^2 p'_8(x) - F(y_* + p_8(x)) - xG(y_* + p_8(x)).$$

At order x^j , the coefficient b_j enters the Taylor expansion of H_8 only through the term $-\mu b_j x^j$; all remaining terms depend only on $b_1, \dots, b_{j-1}$. Since $\mu \neq 0$, the coefficients $b_1, \dots, b_8$ may therefore be chosen recursively so that

$$H_8^{(j)}(0) = 0, \quad 0 \leq j \leq 8.$$

The coefficient of x is $-\mu b_1 - G(y_*)$, which gives the stated value of b_1 . Since $H_8 \in C^9$ on an interval containing the origin, Taylor's theorem applied to $H_8^{(m)}$ gives $H_8^{(m)}(x) = O(x^{9-m})$ for $0 \leq m \leq 4$. Repeated use of $dx/dt = -x^2$ then yields

$$D_8^{(\ell)}(t) = O(t^{-9-\ell}), \quad 0 \leq \ell \leq 4. \quad (\text{B.3})$$

Set $R = Y - P_8$. For all sufficiently large t , both $Y(t)$ and $P_8(t)$ belong to a compact subinterval of I . The mean-value formula gives the exact equation

$$R'(t) = A(t)R(t) - D_8(t), \quad (\text{B.4})$$

where

$$A(t) = \int_0^1 [F'(P_8(t) + \theta R(t)) + t^{-1}G'(P_8(t) + \theta R(t))] d\theta.$$

Since $Y(t), P_8(t) \rightarrow y_*$, one has $A(t) \rightarrow \mu$.

Assume first that $\mu < 0$. Choose $T \geq T_0$ and $\alpha > 0$ such that $A(t) \leq -\alpha$ for $t \geq T$. Variation of constants gives

$$R(t) = \exp \left\{ \int_T^t A(r) dr \right\} R(T) - \int_T^t \exp \left\{ \int_s^t A(r) dr \right\} D_8(s) ds.$$

The first term is exponentially small. For the integral term, split the integral at $t/2$. For $t \geq 2T$, (B.3) gives

$$\int_T^{t/2} e^{-\alpha(t-s)} |D_8(s)| ds \leq e^{-\alpha t/2} \int_T^\infty |D_8(s)| ds = O(e^{-\alpha t/2}).$$

On $[t/2, t]$, one has $s^{-9} \leq 2^9 t^{-9}$, and hence

$$\int_{t/2}^t e^{-\alpha(t-s)} s^{-9} ds \leq 2^9 t^{-9} \int_{t/2}^t e^{-\alpha(t-s)} ds \leq \frac{2^9}{\alpha} t^{-9}.$$

Together with (B.3), these estimates prove $R(t) = O(t^{-9})$.

Assume now that $\mu > 0$. Increase T so that $A(t) \geq \alpha > 0$ for $t \geq T$. Variation of constants again gives

$$R(t) = e^{-\int_t^S A(r) dr} R(S) + \int_t^S e^{-\int_t^s A(r) dr} D_8(s) ds.$$

Since $R(S) \rightarrow 0$, passing to the limit $S \rightarrow \infty$ yields

$$R(t) = \int_t^\infty e^{-\int_t^s A(r) dr} D_8(s) ds. \quad (\text{B.5})$$

The bound $A \geq \alpha$ and (B.3) give $R(t) = O(t^{-9})$.

It remains to control derivatives of R . Suppose inductively that $R^{(j)}(t) = O(t^{-9})$ for $0 \leq j \leq m$, where $m \leq 3$. Differentiation of the displayed formula for A shows that $A^{(j)}(t) = O(1)$ for $0 \leq j \leq m$: every term contains only bounded derivatives of F and G , derivatives of the explicit function P_8 , and derivatives of R already controlled by the induction hypothesis. Differentiating (B.4) m times therefore gives

$$R^{(m+1)}(t) = \sum_{j=0}^m \binom{m}{j} A^{(j)}(t) R^{(m-j)}(t) - D_8^{(m)}(t) = O(t^{-9}).$$

Thus $R^{(\ell)}(t) = O(t^{-9})$ for $0 \leq \ell \leq 4$. Finally,

$$Y - y_* - \sum_{j=1}^4 b_j t^{-j} = \sum_{j=5}^8 b_j t^{-j} + R,$$

and differentiating this identity proves (B.2). □

Lemma B.2. *Let W be a positive solution of (A.1) with parameter $\lambda > 0$.*

If $W(u) \sim u/\lambda$ as $u \downarrow 0$, then W extends to a C^4 function on $[0, 1)$ and

$$W(u) = \frac{u}{\lambda} + (\lambda^{-4} - \lambda^{-1}) u^2 + O(u^3) \quad \text{as } u \downarrow 0.$$

More precisely, there exist $\eta > 0$, coefficients c_3, c_4, c_5 , and a function $\rho_0 \in C^4((0, \eta])$ such that

$$W(u) = \frac{u}{\lambda} + (\lambda^{-4} - \lambda^{-1}) u^2 + \sum_{j=3}^5 c_j u^j + \rho_0(u), \quad \rho_0^{(k)}(u) = O(u^{6-k}) \quad (\text{B.6})$$

for $0 \leq k \leq 4$.

If $W(u) \sim (1-u)/\lambda$ as $u \uparrow 1$, then W extends to a C^4 function on $(0, 1]$ and

$$W(u) = \frac{1-u}{\lambda} - (\lambda^{-4} + \lambda^{-1}) (1-u)^2 + O((1-u)^3) \quad \text{as } u \uparrow 1.$$

More precisely, there exist $\eta > 0$, coefficients $\hat{c}_3, \hat{c}_4, \hat{c}_5$, and a function $\rho_1 \in C^4((0, \eta])$ such that, with $r = 1 - u$,

$$W(1-r) = \frac{r}{\lambda} - (\lambda^{-4} + \lambda^{-1}) r^2 + \sum_{j=3}^5 \hat{c}_j r^j + \rho_1(r), \quad \rho_1^{(k)}(r) = O(r^{6-k}) \quad (\text{B.7})$$

for $0 \leq k \leq 4$.

At each endpoint for which the corresponding linear asymptotic holds, there exist $\eta > 0$ and $C < \infty$ such that

$$|W'(u)| + |W(u)W''(u)| + |W(u)^2W'''(u)| \leq C \quad (\text{B.8})$$

throughout the associated one-sided interval $(0, \eta]$ or $[1 - \eta, 1)$.

Proof. Consider first the endpoint $u = 0$ and set

$$Y(t) = tW(t^{-1}), \quad t > 1.$$

Substitution of $W(u) = uY(u^{-1})$ into (A.1) gives

$$Y'(t) = -\frac{\lambda Y(t) - 1}{Y(t)^2} - \frac{1 - Y(t)^3}{tY(t)^2}, \quad Y(t) \longrightarrow \lambda^{-1}.$$

Thus Lemma B.1 applies with

$$F_0(y) = \frac{1 - \lambda y}{y^2}, \quad G_0(y) = y - y^{-2}, \quad y_* = \lambda^{-1}.$$

Since

$$F'_0(\lambda^{-1}) = -\lambda^3, \quad -\frac{G_0(\lambda^{-1})}{F'_0(\lambda^{-1})} = \lambda^{-4} - \lambda^{-1},$$

there are coefficients b_2, b_3, b_4 and a remainder R_0 such that $R_0^{(k)}(t) = O(t^{-5-k})$, $0 \leq k \leq 4$, and

$$Y(t) = \lambda^{-1} + (\lambda^{-4} - \lambda^{-1})t^{-1} + \sum_{j=2}^4 b_j t^{-j} + R_0(t).$$

Multiplication by $u = t^{-1}$ gives (B.6), once the transformed remainder is shown to have the stated derivative bounds.

For the endpoint $u = 1$, set

$$\hat{Y}(t) = tW(1 - t^{-1}), \quad t > 1.$$

Then

$$\hat{Y}'(t) = \frac{\lambda \hat{Y}(t) - 1}{\hat{Y}(t)^2} + \frac{1 + \hat{Y}(t)^3}{t \hat{Y}(t)^2}, \quad \hat{Y}(t) \longrightarrow \lambda^{-1}.$$

Here Lemma B.1 applies with

$$F_1(y) = \frac{\lambda y - 1}{y^2}, \quad G_1(y) = y + y^{-2}.$$

Moreover,

$$F'_1(\lambda^{-1}) = \lambda^3, \quad -\frac{G_1(\lambda^{-1})}{F'_1(\lambda^{-1})} = -(\lambda^{-4} + \lambda^{-1}).$$

The case $F'_1(\lambda^{-1}) > 0$ is covered by the terminal representation (B.5). It yields (B.7) after multiplication by $r = t^{-1} = 1 - u$.

We record explicitly the change of variables used for both remainders. Let R be a function satisfying

$$\frac{d^k R}{dt^k}(t) = O(t^{-5-k}), \quad 0 \leq k \leq 4,$$

and define $E(r) = rR(r^{-1})$ for $r > 0$. Put $t = r^{-1}$. Derivatives of E in the following display are taken with respect to r , whereas derivatives of R are taken with respect to t . Since $dt/dr = -t^2$, the chain rule gives

$$\begin{aligned} \frac{dE}{dr}(r) &= R(t) - t \frac{dR}{dt}(t), & \frac{d^2 E}{dr^2}(r) &= t^3 \frac{d^2 R}{dt^2}(t), \\ \frac{d^3 E}{dr^3}(r) &= -3t^4 \frac{d^2 R}{dt^2}(t) - t^5 \frac{d^3 R}{dt^3}(t), & \frac{d^4 E}{dr^4}(r) &= 12t^5 \frac{d^2 R}{dt^2}(t) + 8t^6 \frac{d^3 R}{dt^3}(t) + t^7 \frac{d^4 R}{dt^4}(t). \end{aligned}$$

Consequently,

$$\frac{d^k E}{dr^k}(r) = O(r^{6-k}), \quad 0 \leq k \leq 4.$$

Thus the transformed remainder, together with its first four derivatives, extends continuously to $r = 0$ with value zero. Taking $r = u$ at the left endpoint and $r = 1 - u$ at the right endpoint proves the asserted C^4 extensions. The expansions then imply (B.8) on the corresponding one-sided intervals. $\square$

Proof of Proposition 4.2. By Proposition A.7 and Lemma A.5, both endpoint branches of W_λ are linear. Lemma B.2 therefore applies at $u = 0$ and $u = 1$.

Define

$$\Theta_\lambda(v) = \int_{1/2}^v \frac{ds}{\beta W_\lambda(1-s)}, \quad 0 < v < 1.$$

As $s \downarrow 0$, $W_\lambda(1-s) \sim s/\lambda$, whereas as $s \uparrow 1$, $W_\lambda(1-s) \sim (1-s)/\lambda$. Hence

$$\Theta_\lambda(v) \longrightarrow -\infty \quad \text{as } v \downarrow 0, \quad \Theta_\lambda(v) \longrightarrow \infty \quad \text{as } v \uparrow 1.$$

Since $\Theta'_\lambda(v) > 0$, the map Θ_λ is a smooth bijection from $(0, 1)$ onto $\mathbb{R}$. Write $\Phi = \Phi_\lambda := \Theta_\lambda^{-1}$. Then Φ satisfies (4.2). Conversely, separation of variables applied to the differential equation in (4.2) shows that every increasing solution satisfies $\Theta_\lambda(\Phi(z)) = z - z_0$ for some constant z_0 . Thus any two such solutions differ by a translation, and the normalization $\Phi(0) = 1/2$ fixes that translation.

Put $q = q_\lambda = \Phi'$ and $u = 1 - \Phi(z)$. Then $q = \beta W_\lambda(u)$ and $u' = -\beta W_\lambda(u)$. In particular,

$$q' = -\beta^2 W_\lambda(u) W'_\lambda(u).$$

Using $a\beta^3 = 2$ and (A.1), one obtains

$$a\Phi'\Phi'' = -2W_\lambda(u)^2 W'_\lambda(u) = -2\lambda W_\lambda(u) + 2u(1-u) = -\kappa_\lambda q + 2\Phi(1-\Phi),$$

which proves (4.3).

All occurrences of W_λ and its derivatives in the following identities are evaluated at $u = 1 - \Phi(z)$. Repeated differentiation gives

$$\frac{q'}{q} = -\beta W'_\lambda, \quad \frac{q''}{q} = \beta^2 ((W'_\lambda)^2 + W_\lambda W''_\lambda), \quad \frac{q'''}{q} = -\beta^3 ((W'_\lambda)^3 + 4W_\lambda W'_\lambda W''_\lambda + W_\lambda^2 W'''_\lambda).$$

The bounds in (B.8) control these expressions on one-sided intervals at 0 and 1. On the complementary compact subinterval of $(0, 1)$ they are bounded by continuity. This proves (4.4).

As $z \rightarrow -\infty$, one has $\Phi(z) \rightarrow 0$ and $u = 1 - \Phi(z) \rightarrow 1$. The linear expansion at $u = 1$ therefore gives

$$\frac{q(z)}{\Phi(z)} = \beta \frac{W_\lambda(1 - \Phi(z))}{\Phi(z)} \longrightarrow \frac{\beta}{\lambda}.$$

Similarly, as $z \rightarrow \infty$, the linear expansion at $u = 0$ gives

$$\frac{q(z)}{1 - \Phi(z)} = \beta \frac{W_\lambda(u)}{u} \longrightarrow \frac{\beta}{\lambda}.$$

This proves (4.5).

Choose $L > 0$ such that both ratios in (4.5) are at least $c := \beta/(2\lambda)$ on their respective tails. Then

$$(\log \Phi)' = \frac{q}{\Phi} \geq c \quad \text{for } z \leq -L, \quad -(\log(1 - \Phi))' = \frac{q}{1 - \Phi} \geq c \quad \text{for } z \geq L.$$

Integration gives the stated exponential bounds after increasing the multiplicative constant on the compact interval $[-L, L]$. Finally, the tail-integral identity

$$\int_{\mathbb{R}} |x| \, d\Phi(x) = \int_{-\infty}^0 \Phi(x) \, dx + \int_0^\infty (1 - \Phi(x)) \, dx$$

shows that the absolute first moment is finite. □

Proof of Proposition 4.3. Proposition A.4 gives $W_\lambda(0) > 0$, and Lemma A.5 gives $W_\lambda(u) \sim (1-u)/\lambda$ as $u \uparrow 1$. Define

$$z(v) = \int_0^v \frac{ds}{\beta W_\lambda(s)}, \quad 0 \leq v < 1.$$

The integrand is continuous at $v = 0$, while the linear expansion at $v = 1$ implies $z(v) \rightarrow \infty$ as $v \uparrow 1$. Since $z'(v) > 0$, the map z is a bijection from $[0, 1)$ onto $[0, \infty)$. Its inverse is the function Ψ_λ in (4.6). Since W_λ is smooth and strictly positive on $[0, 1)$, with $W_\lambda(0) > 0$, the one-sided inverse function theorem shows that Ψ_λ is smooth on $[0, \infty)$. Separation of variables gives uniqueness.

For $z \geq 0$, write $v = \Psi_\lambda(z)$. Then $q_\lambda = \beta W_\lambda(v)$ and

$$q'_\lambda = \beta^2 W_\lambda(v) W'_\lambda(v).$$

Since $a\beta^3 = 2$, equation (A.1) gives

$$aq_\lambda q'_\lambda = 2W_\lambda(v)^2 W'_\lambda(v) = 2\lambda W_\lambda(v) - 2v(1-v) = \kappa_\lambda q_\lambda - 2\Psi_\lambda(1 - \Psi_\lambda),$$

which proves (4.7), including the value at $z = 0$ by continuity.

All occurrences of W_λ and its derivatives below are evaluated at $v = \Psi_\lambda(z)$. Repeated differentiation gives

$$\frac{q'_\lambda}{q_\lambda} = \beta W'_\lambda, \quad \frac{q''_\lambda}{q_\lambda} = \beta^2 ((W'_\lambda)^2 + W_\lambda W''_\lambda), \quad \frac{q'''_\lambda}{q_\lambda} = \beta^3 ((W'_\lambda)^3 + 4W_\lambda W'_\lambda W''_\lambda + W_\lambda^2 W'''_\lambda).$$

On an interval $[0, \eta]$, Proposition A.4 implies that W_λ is smooth and bounded away from zero. On an interval $[1-\eta, 1)$, Lemma B.2 gives (B.8). Continuity controls the remaining compact interval. Hence (4.8) follows. The linear expansion at $v = 1$ also gives (4.10).

Choose $L > 0$ such that $q_\lambda(z)/(1 - \Psi_\lambda(z)) \geq c := \beta/(2\lambda)$ for $z \geq L$. Then

$$-\frac{d}{dz} \log(1 - \Psi_\lambda(z)) = \frac{q_\lambda(z)}{1 - \Psi_\lambda(z)} \geq c, \quad z \geq L.$$

Integration gives the stated exponential bound. Since the distribution is supported on $[0, \infty)$,

$$\int_{[0, \infty)} x \, d\Psi_\lambda(x) = \int_0^\infty (1 - \Psi_\lambda(x)) \, dx < \infty.$$

It remains to construct the extension of q_λ . Set $h = \log q_\lambda$ on $[0, \infty)$. From (4.8),

$$h' = \frac{q'_\lambda}{q_\lambda}, \quad h'' = \frac{q''_\lambda}{q_\lambda} - \left(\frac{q'_\lambda}{q_\lambda}\right)^2, \quad h''' = \frac{q'''_\lambda}{q_\lambda} - 3\frac{q'_\lambda q''_\lambda}{q_\lambda^2} + 2\left(\frac{q'_\lambda}{q_\lambda}\right)^3$$

are bounded on $[0, \infty)$.

Let

$$P(z) = \sum_{j=0}^3 \frac{h^{(j)}(0)}{j!} z^j,$$

and choose $\chi \in C^\infty(\mathbb{R})$ such that $\chi = 1$ on $[-1/2, 0]$ and $\chi = 0$ on $(-\infty, -1]$. Define

$$\bar{h}(z) = \begin{cases} h(z), & z \geq 0, \\ \chi(z)P(z) + (1 - \chi(z))h(0), & z < 0. \end{cases}$$

Then $\bar{h} \in C^3(\mathbb{R})$, it agrees with h on $[0, \infty)$, and its first three derivatives are bounded. Set $\bar{q}_\lambda = e^{\bar{h}}$. The identities

$$\frac{\bar{q}'_\lambda}{\bar{q}_\lambda} = \bar{h}', \quad \frac{\bar{q}''_\lambda}{\bar{q}_\lambda} = \bar{h}'' + (\bar{h}')^2, \quad \frac{\bar{q}'''_\lambda}{\bar{q}_\lambda} = \bar{h}''' + 3\bar{h}'\bar{h}'' + (\bar{h}')^3$$

prove (4.9). □

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Data availability

No empirical data sets were generated or analyzed in this study. The machine-checkable Lean sources, manuscript-formalization crosswalks, and verification reports are included in the versioned software release described below.

Code availability

The Lean 4 development formalizing the appendix, the traditional graph semantics, and the proof graph for the three principal results is versioned at <https://github.com/iyalice/Series-Parallel>. Its frozen release is v1.0.1, at commit a661859343939016608333bd2673d433c34518f2. The release landing page is <https://github.com/iyalice/Series-Parallel/releases/tag/v1.0.1>, and the corresponding source archive is <https://github.com/iyalice/Series-Parallel/archive/refs/tags/v1.0.1.zip>. The release contains the manuscript-formalization source crosswalk, build and axiom reports, reproduction instructions, and the explicitly documented external mathematical interfaces that form the stated trust boundary. It is checked with Lean 4.32.1 and mathlib v4.32.1 at commit 520045ab14e26149ee970e2e617ca04b09bde5d6, and it is distributed under the Apache-2.0 license. The repository is currently access-controlled, and no DOI has been assigned to this release. The software citation is [9].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT with the GPT-5.6 model identified in the project record and OpenAI Codex with a GPT-5-family model (the client did not expose a more specific deployment identifier). These tools assisted with initial mathematical proof sketches, Lean declaration and proof drafts, mathlib search, consistency checks, language editing, and suggestions for the manuscript-formalization crosswalk. The authors subsequently reconstructed, checked, and revised the mathematical arguments; verified the references; reviewed the Lean source and its documented axiom dependencies; and validated the manuscript-formalization correspondence. The authors made all final mathematical and editorial decisions and take full responsibility for the content of the article.

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